

# Cross-Wearer Gesture Recognition — LOUO Variant Comparison

Generated 2026-05-14 07:24:47

## Data & configuration

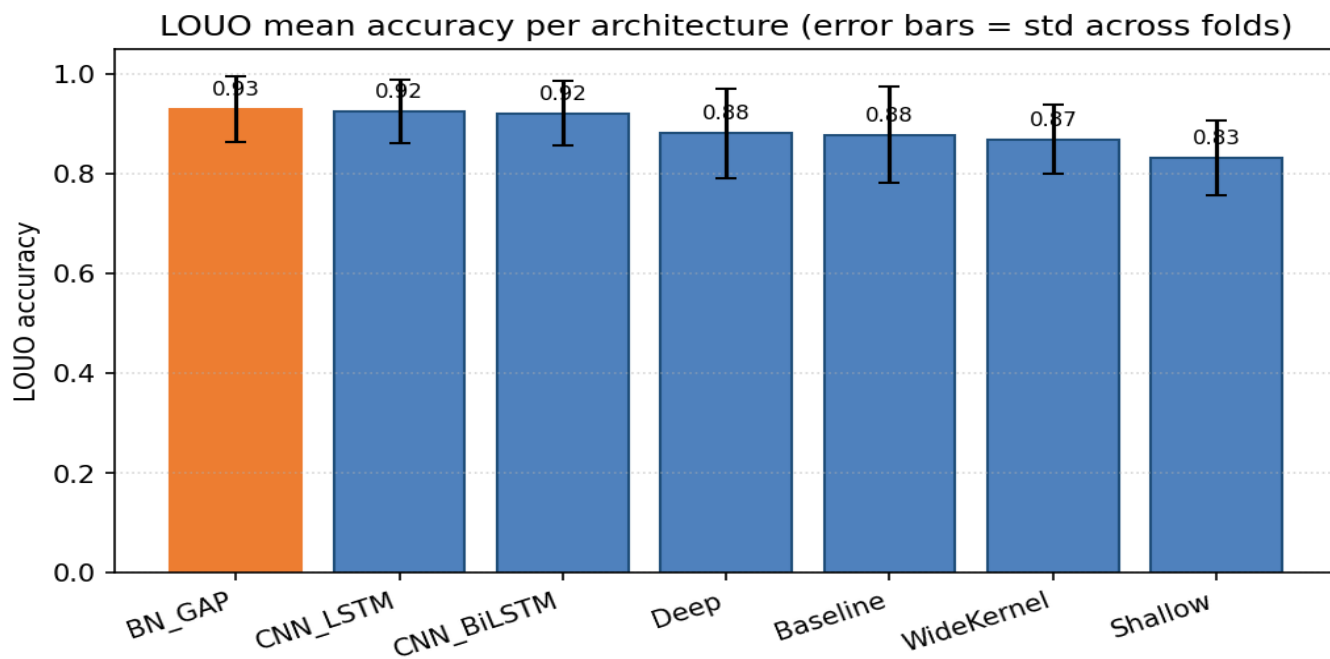
|                             |                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| LOUO path 1                 | /home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic                                                                                     |
| LOUO path 2                 | /home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic                                                                                     |
| LOUO path 3                 | /home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic                                                                                   |
| LOUO path 4                 | /home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic                                                                                   |
| LOUO path 5                 | /home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic                                                                                    |
| LOUO path 6                 | /home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic                                                                                   |
| LOUO path 7                 | /home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic                                                                                   |
| LOUO path 8                 | /home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic                                                                                  |
| LOUO path 9                 | /home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic                                                                               |
| LOUO path 10                | /home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic                                                                                    |
| LOUO path 11                | /home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Dynamic                                                                                   |
| LOUO path 12                | /home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic                                                                                  |
| LOUO path 13                | /home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic                                                                               |
| LOUO users loaded           | 13 (11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry, 8_Jestin) |
| Excluded classes            | — (none)                                                                                                                                  |
| Classes                     | 7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll                               |
| Sensor channels             | 176 (sequence length 30)                                                                                                                  |
| Resample / filter           | 30 steps · Butterworth cutoff 10.0 Hz, order 4                                                                                            |
| Per-trial normalisation     | zscore                                                                                                                                    |
| Downstream normalisation    | minmax                                                                                                                                    |
| Sampling rate               | 30.0 Hz                                                                                                                                   |
| Random state                | 42                                                                                                                                        |
| Augmentation                | off                                                                                                                                       |
| Epochs · batch size         | 40 · 16                                                                                                                                   |
| Early stopping              | on · monitor val_loss · patience=8                                                                                                        |
| LOUO inner-validation       | user                                                                                                                                      |
| LOUO variants               | Baseline, Shallow, Deep, BN_GAP, WideKernel, CNN_LSTM, CNN_BiLSTM                                                                         |
| Final-retrain variant       | BN_GAP_Wide                                                                                                                               |
| Final-retrain held-out user | — (trained on all users)                                                                                                                  |

## Variant comparison (LOUO)

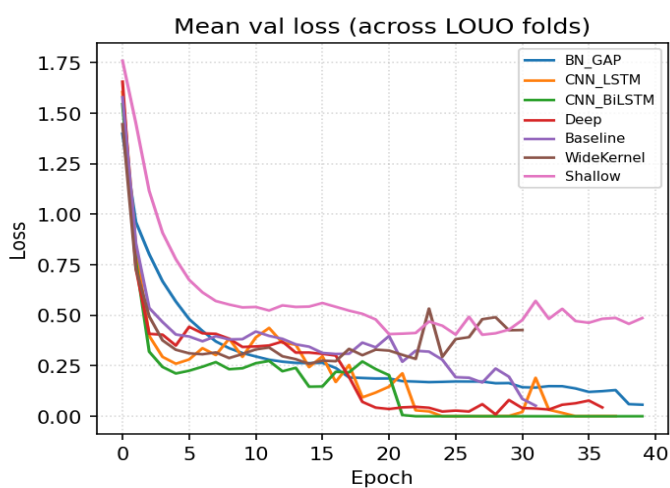
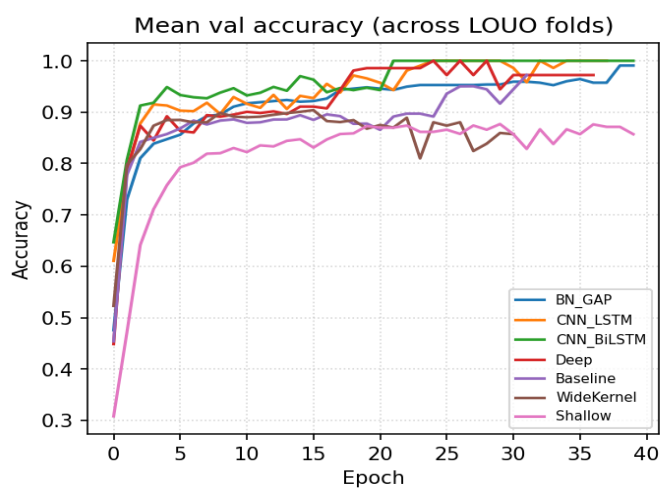
| Variant    | Params  | LOUO mean | LOUO std | Folds | Held-out user acc |
|------------|---------|-----------|----------|-------|-------------------|
| BN_GAP     | 57,319  | 0.9294    | 0.0663   | 13    | —                 |
| CNN_LSTM   | 68,455  | 0.9248    | 0.0633   | 13    | —                 |
| CNN_BiLSTM | 105,575 | 0.9207    | 0.0657   | 13    | —                 |
| Deep       | 72,935  | 0.8802    | 0.0902   | 13    | —                 |
| Baseline   | 51,815  | 0.8771    | 0.0968   | 13    | —                 |
| WideKernel | 70,247  | 0.8683    | 0.0690   | 13    | —                 |
| Shallow    | 41,767  | 0.8311    | 0.0742   | 13    | —                 |

Best LOUO mean: <b>BN\_GAP</b>

## LOUO accuracy per variant

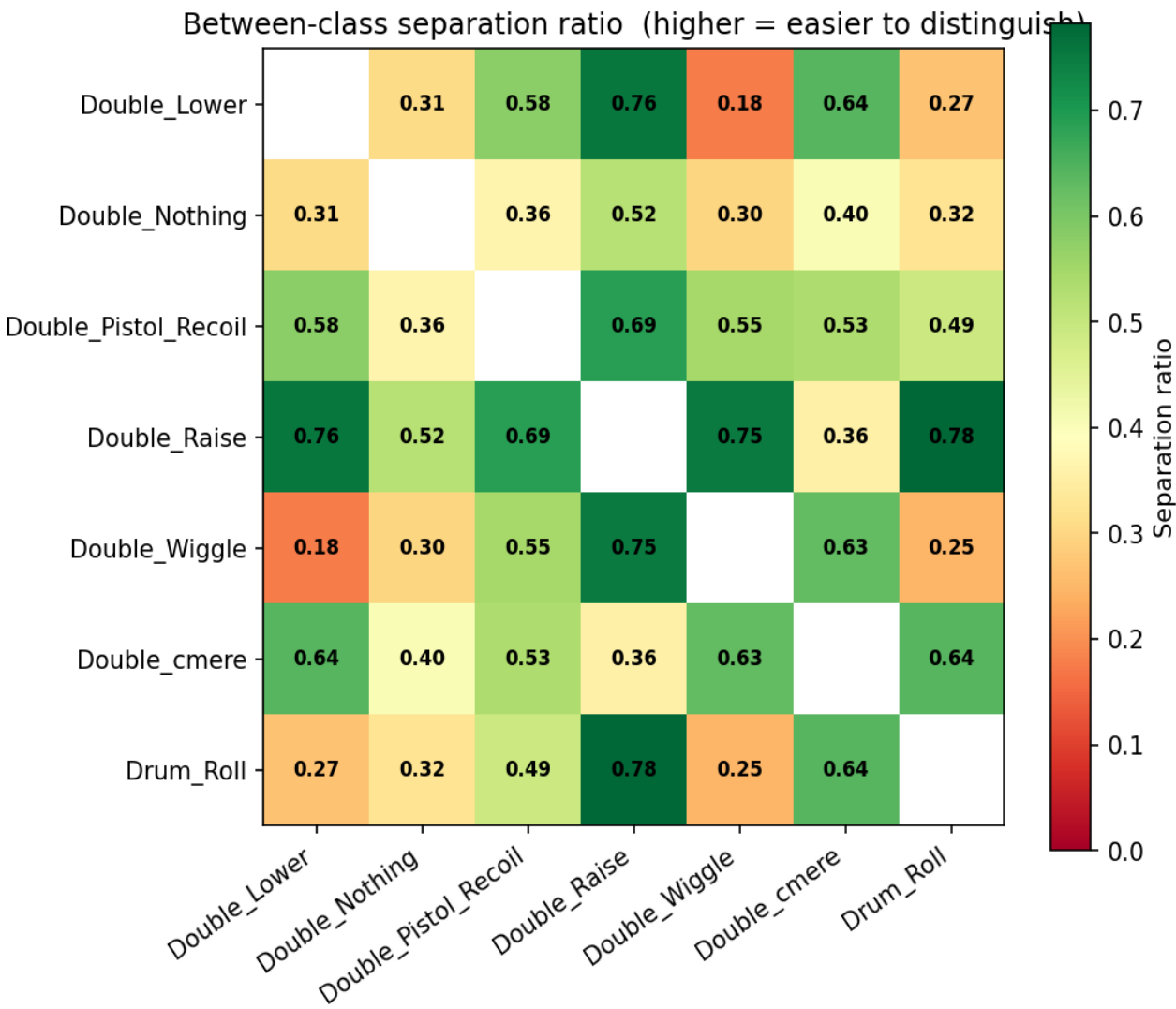
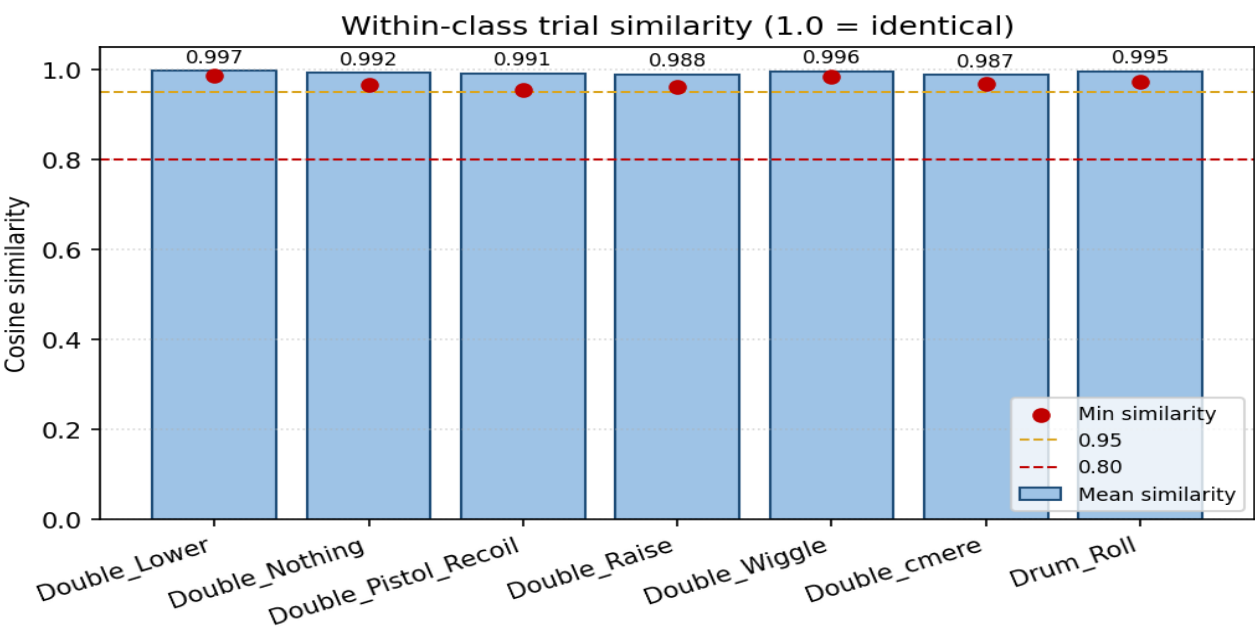


### Mean training curves across LOUO folds

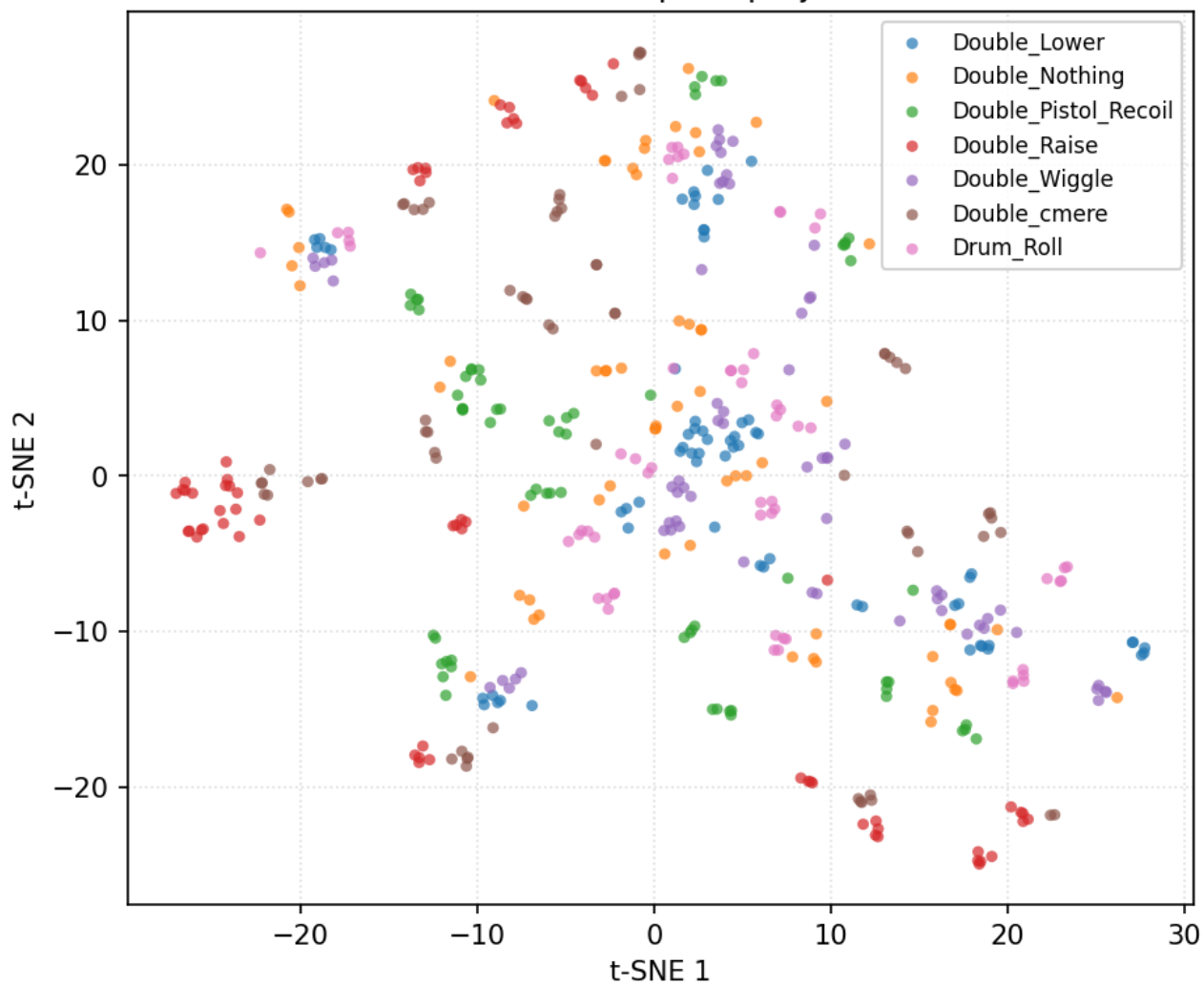


# Data diagnostics — pooled across users

Filtered + resampled trials from every LOUO user (before per-trial normalisation).



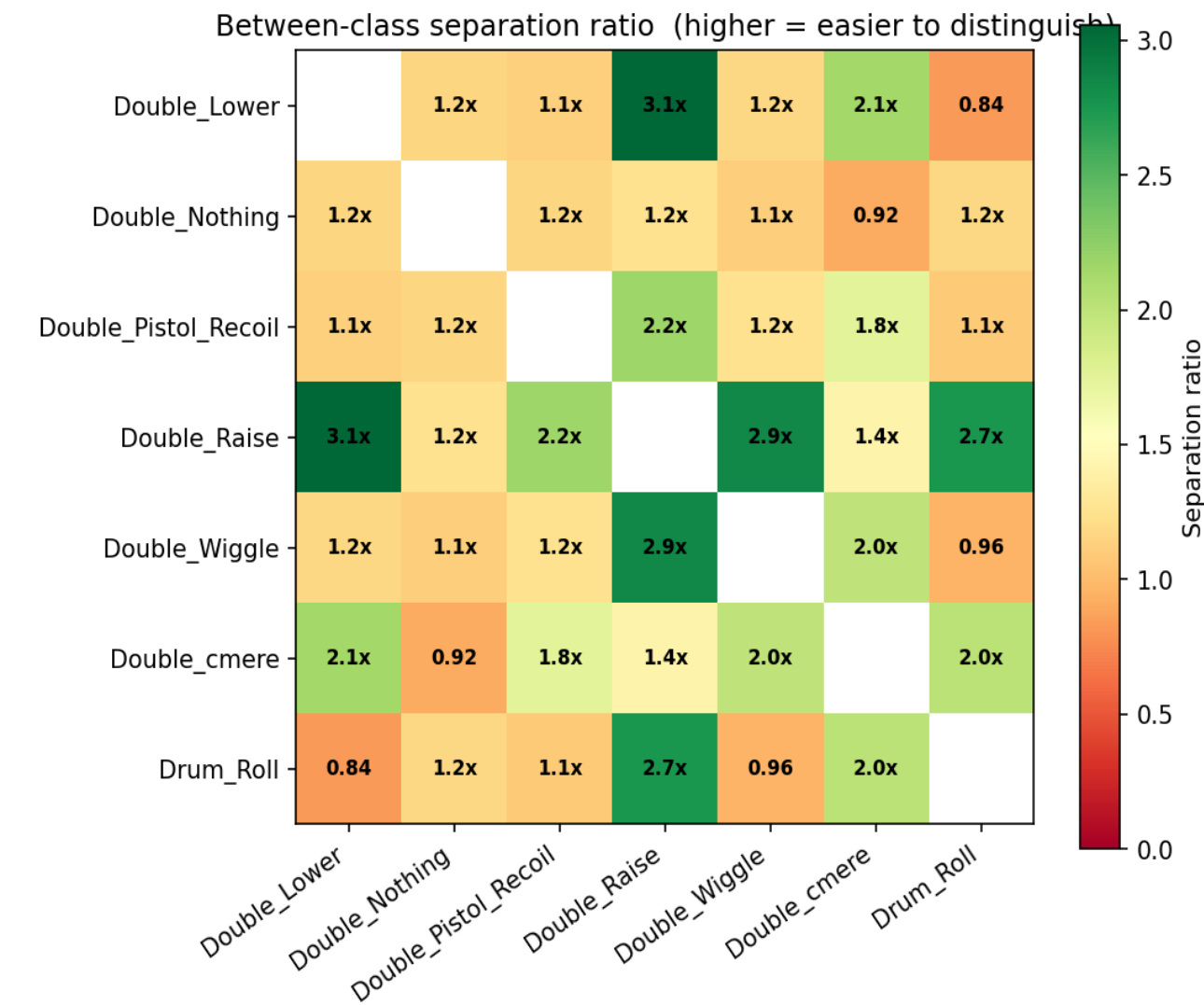
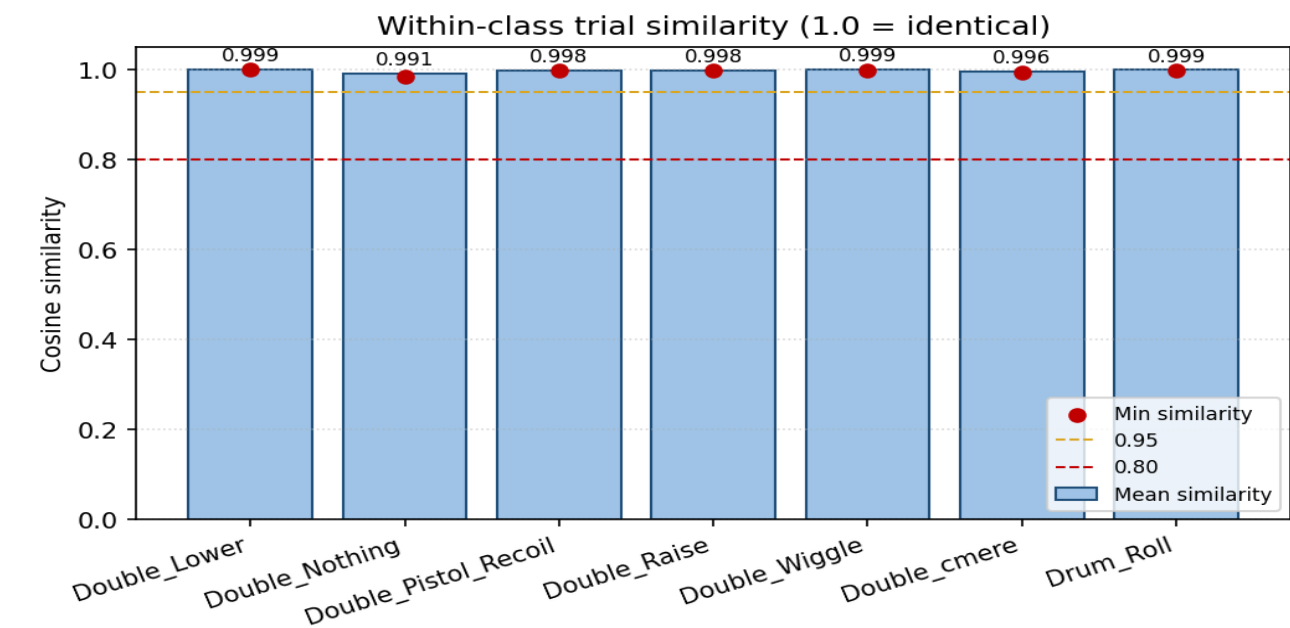
t-SNE feature space projection



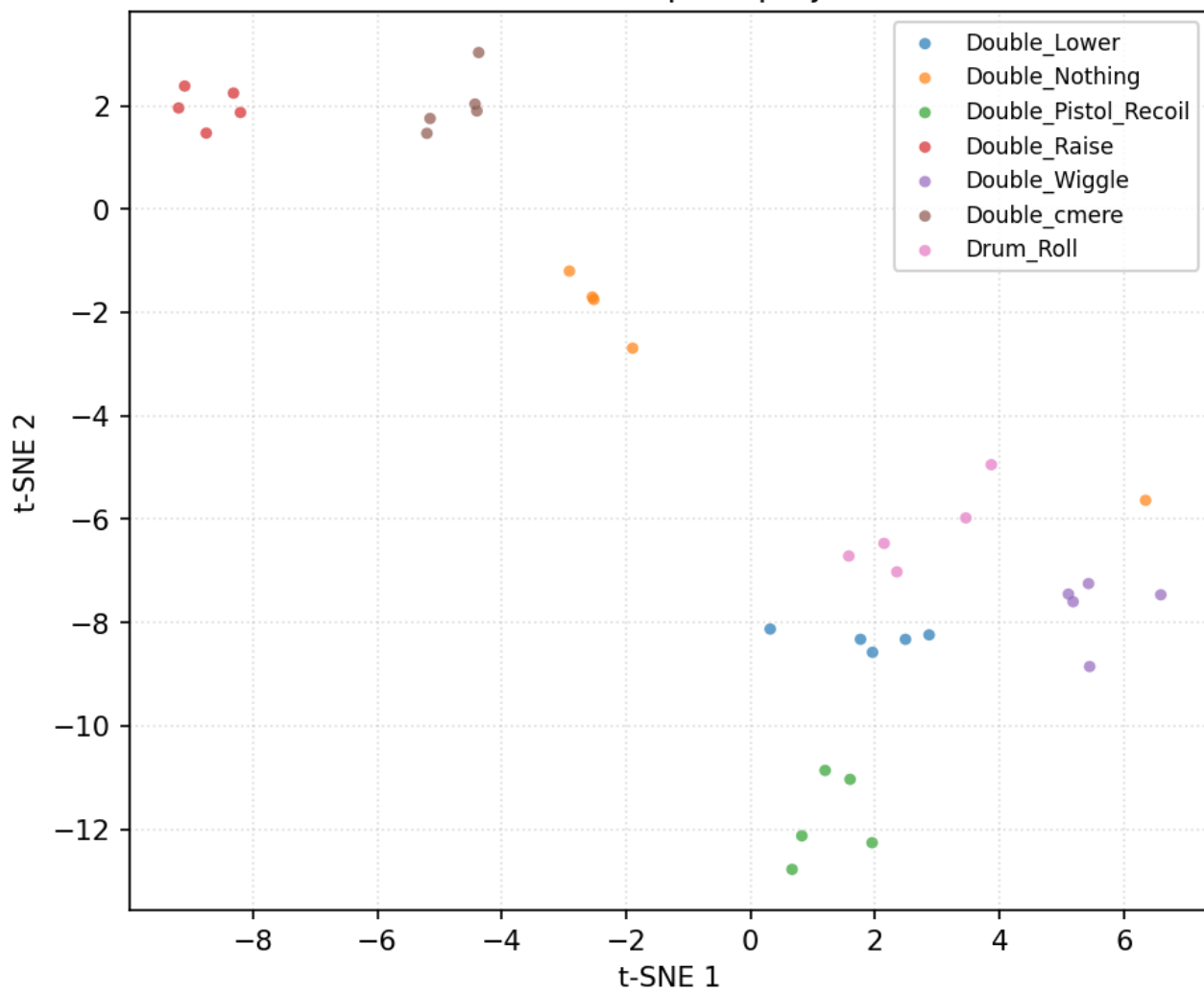
Reading guide: within-class similarity > 0.95 = very consistent; < 0.80 = high variance. Separation ratio > 100x = trivially separable, 20-100x = easy, 5-20x = moderate, < 5x = hard (expect confusion).

# Data diagnostics — user 11\_Mansh

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

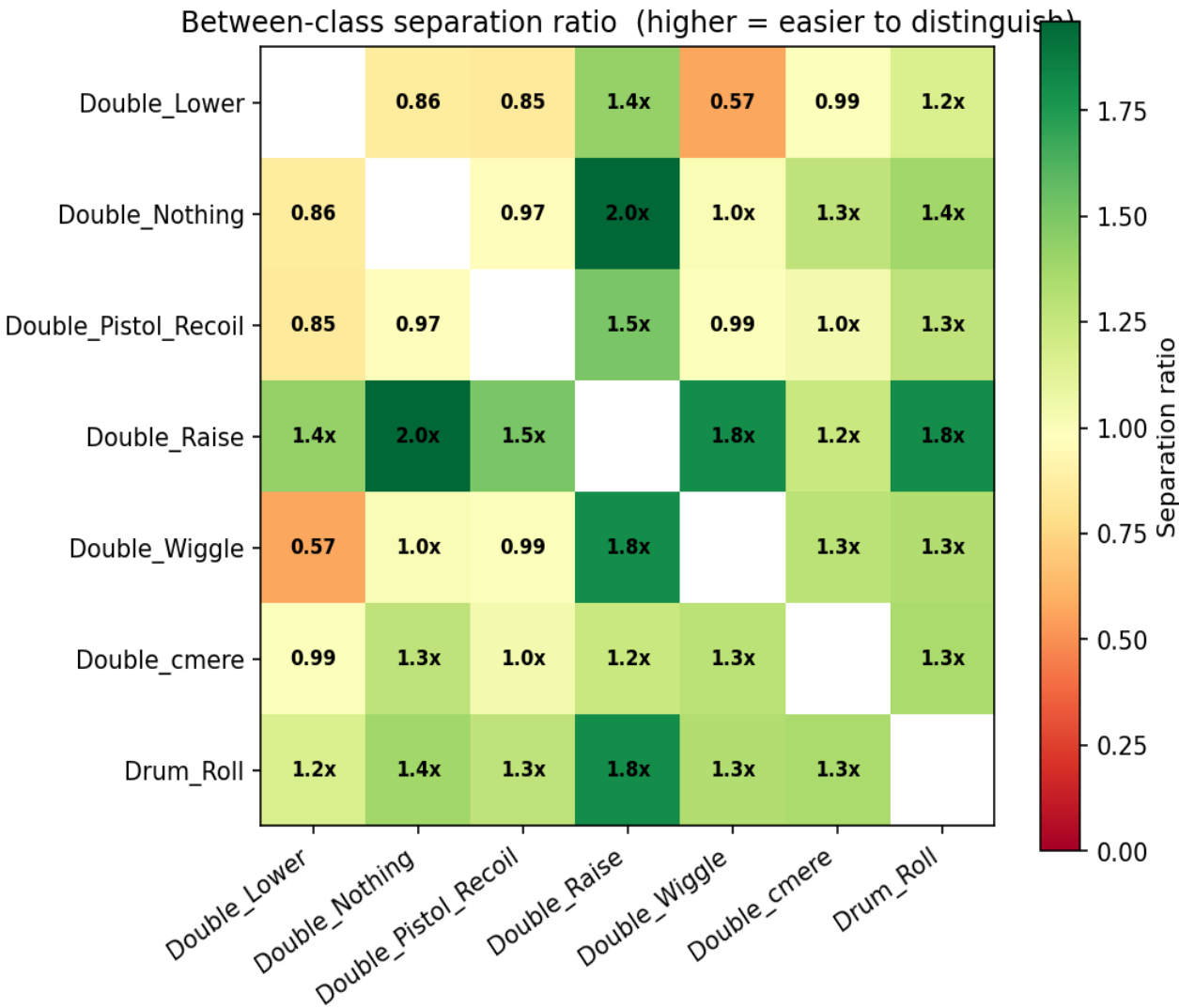
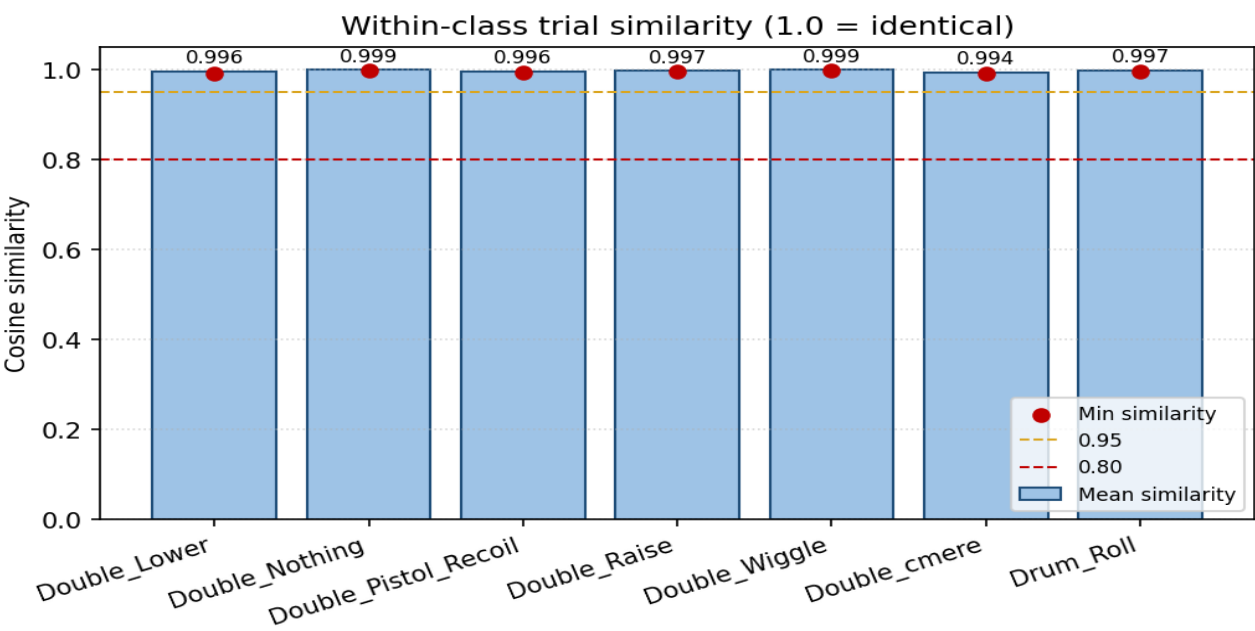


t-SNE feature space projection

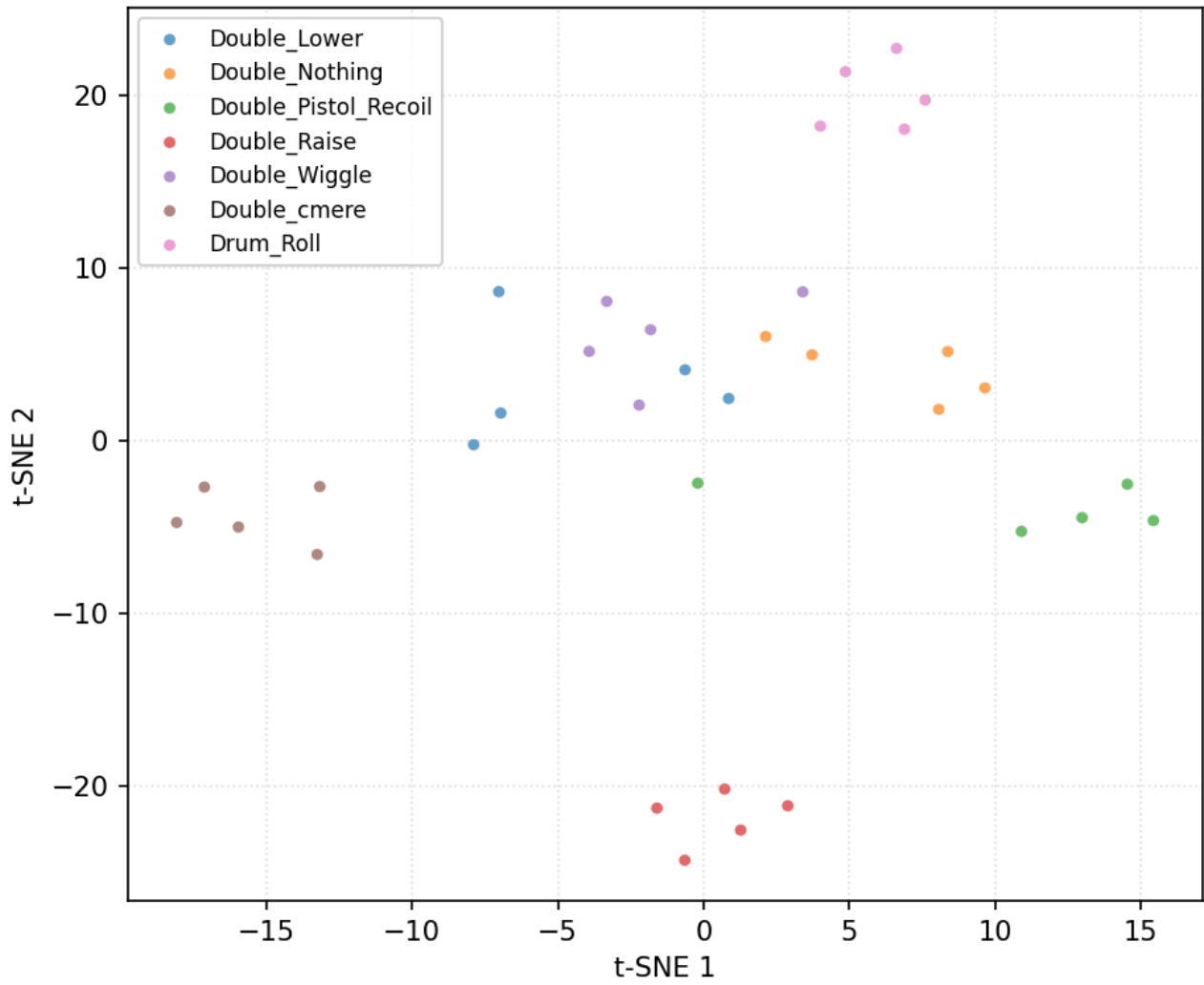


# Data diagnostics — user 12\_Marcus

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).



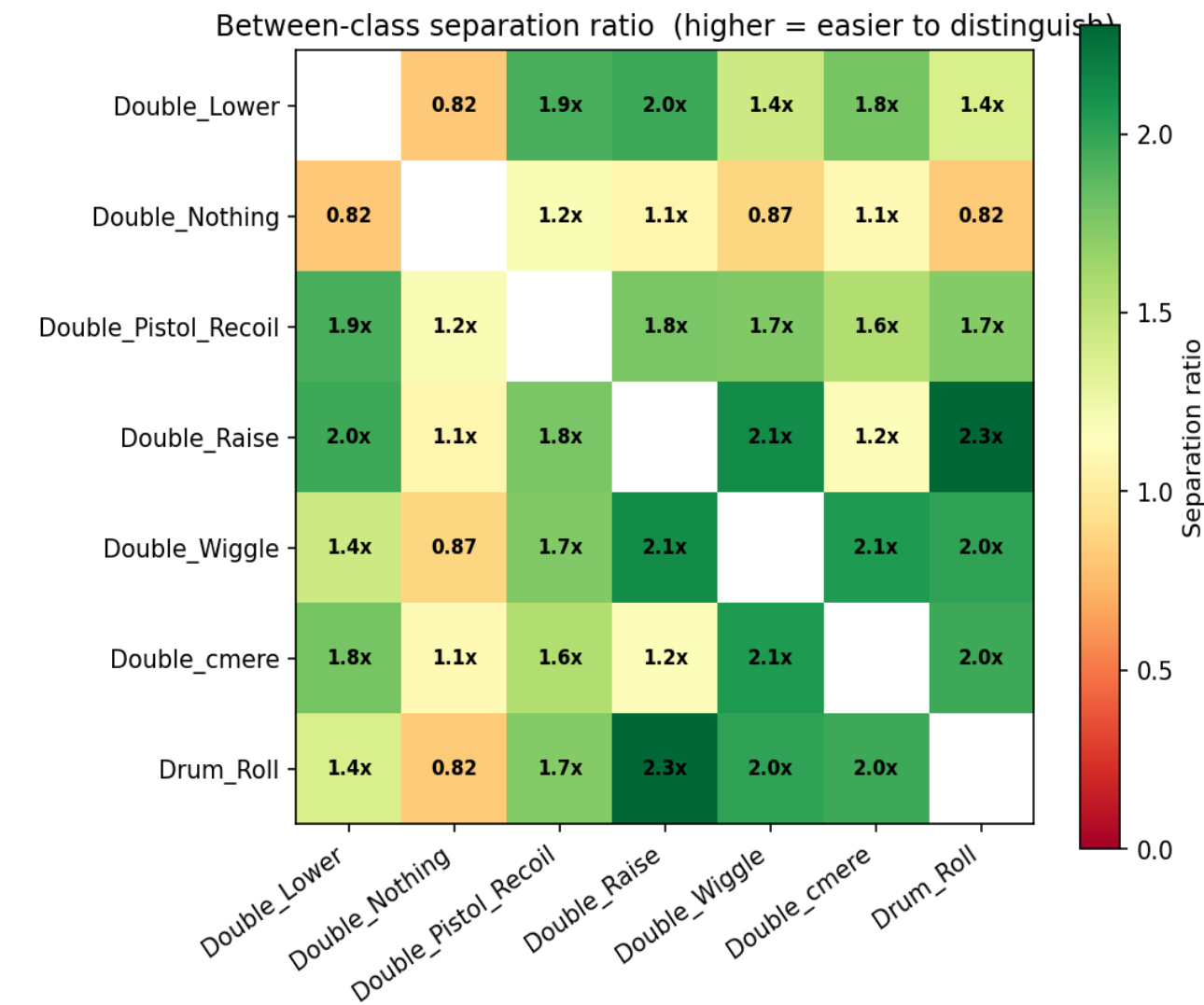
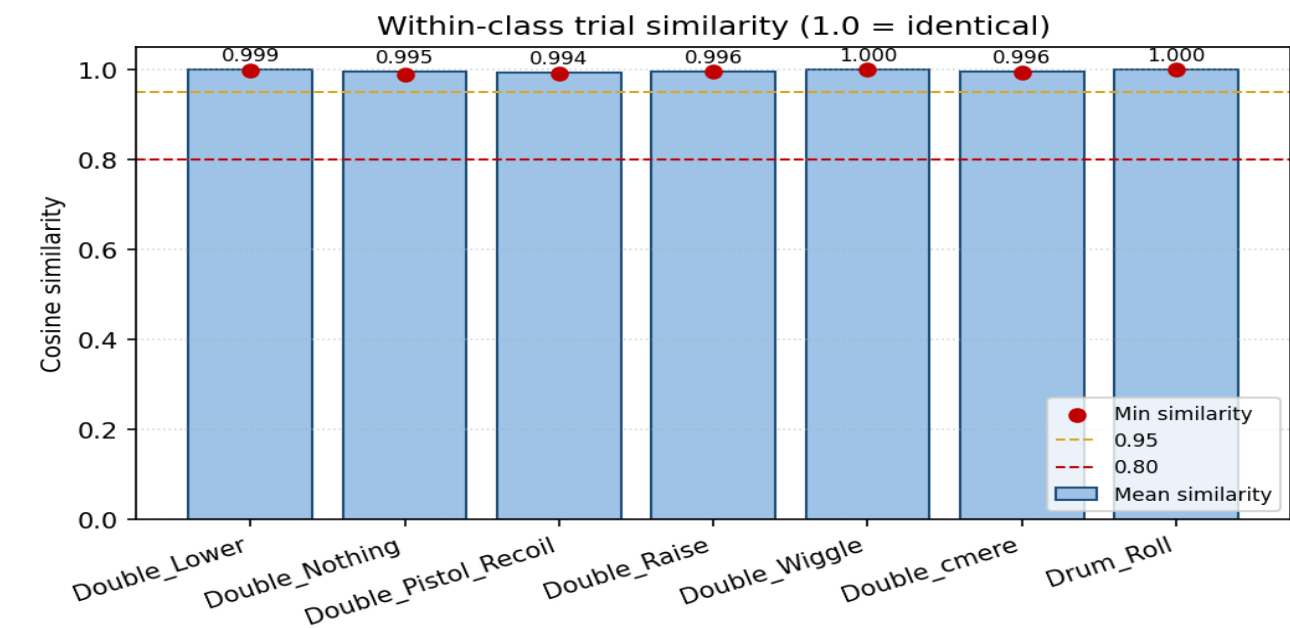
t-SNE feature space projection



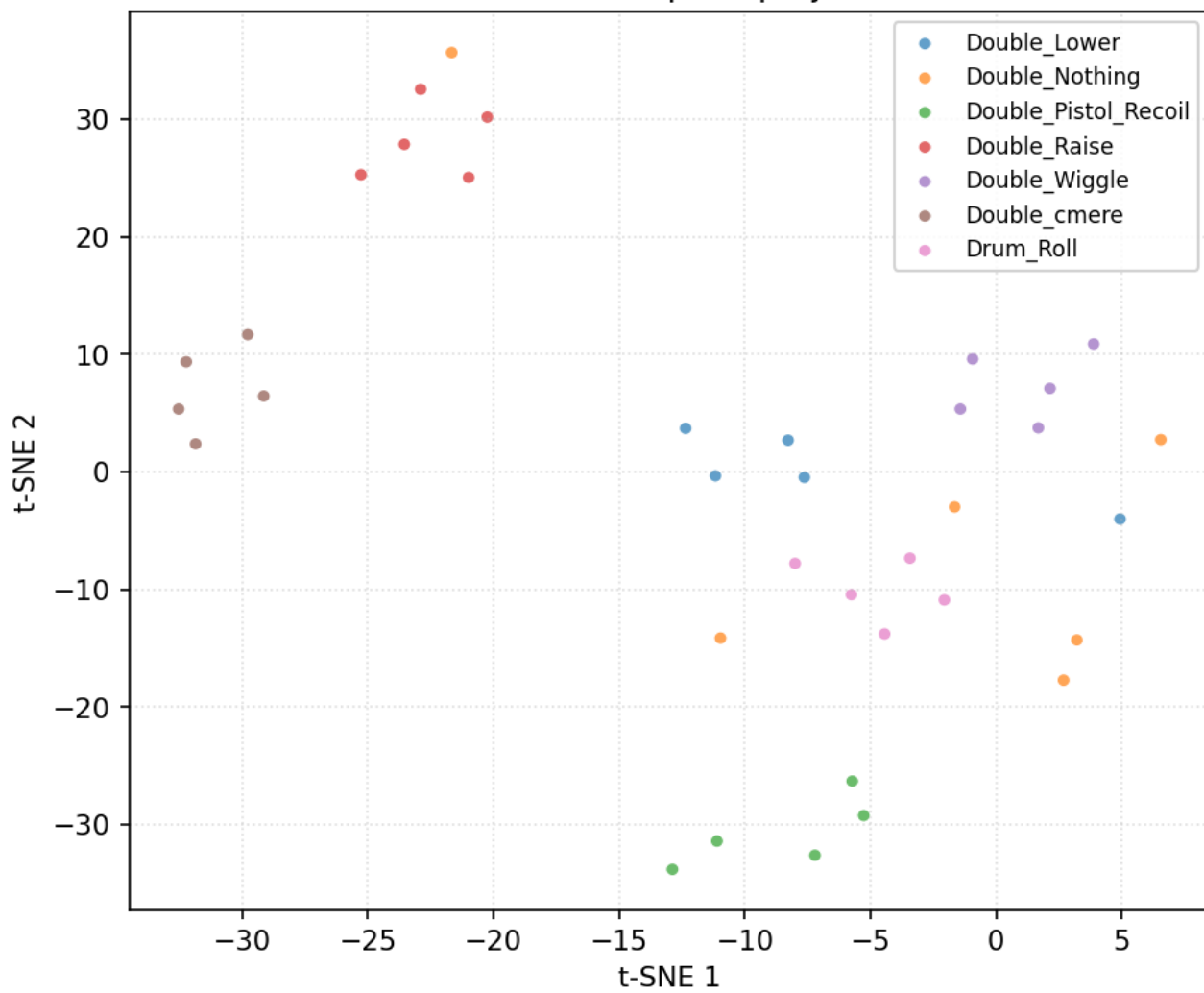


# Data diagnostics — user 14\_StephenV2

36 trials, 7 classes (filtered + resampled, before per-trial normalisation).

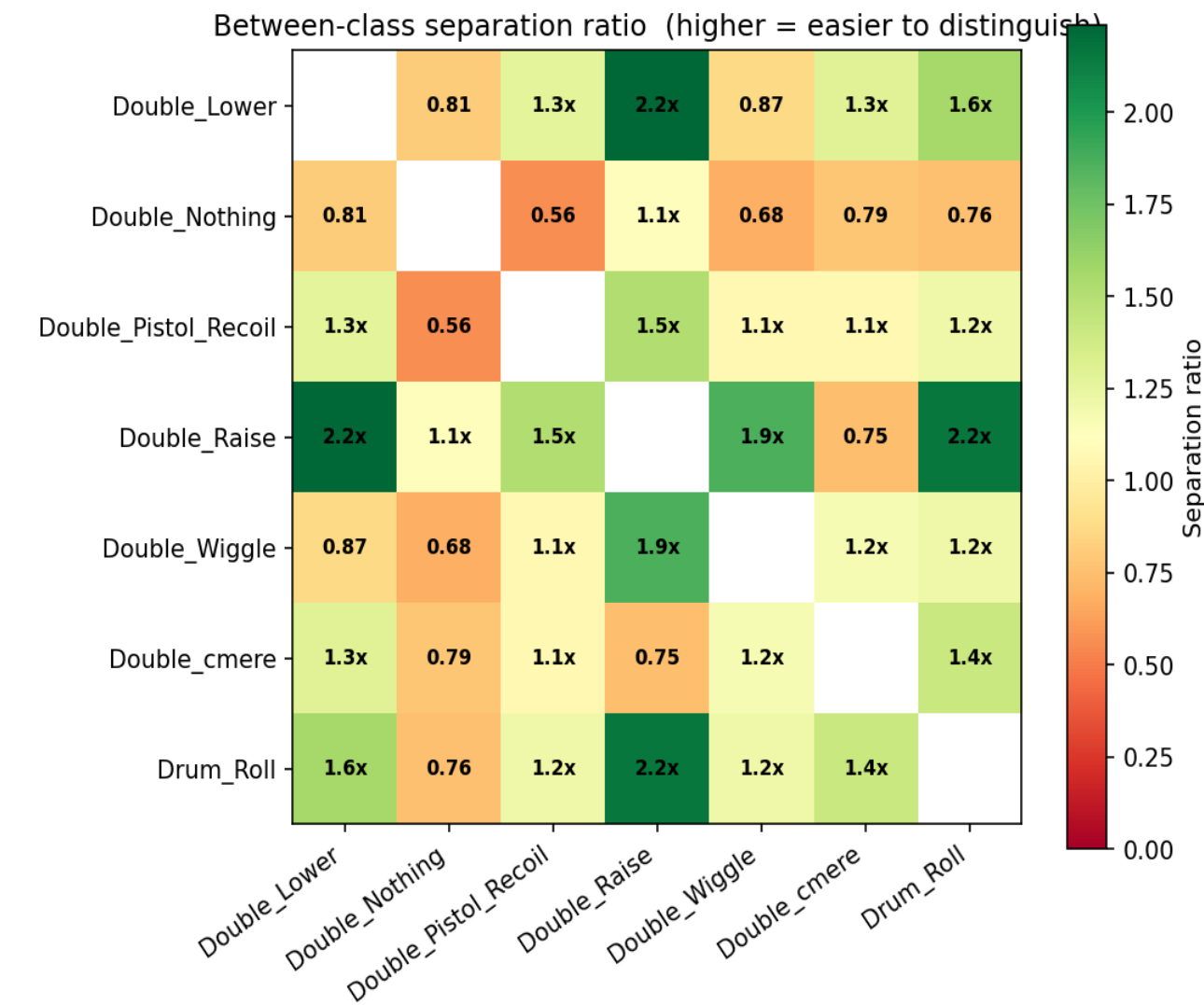
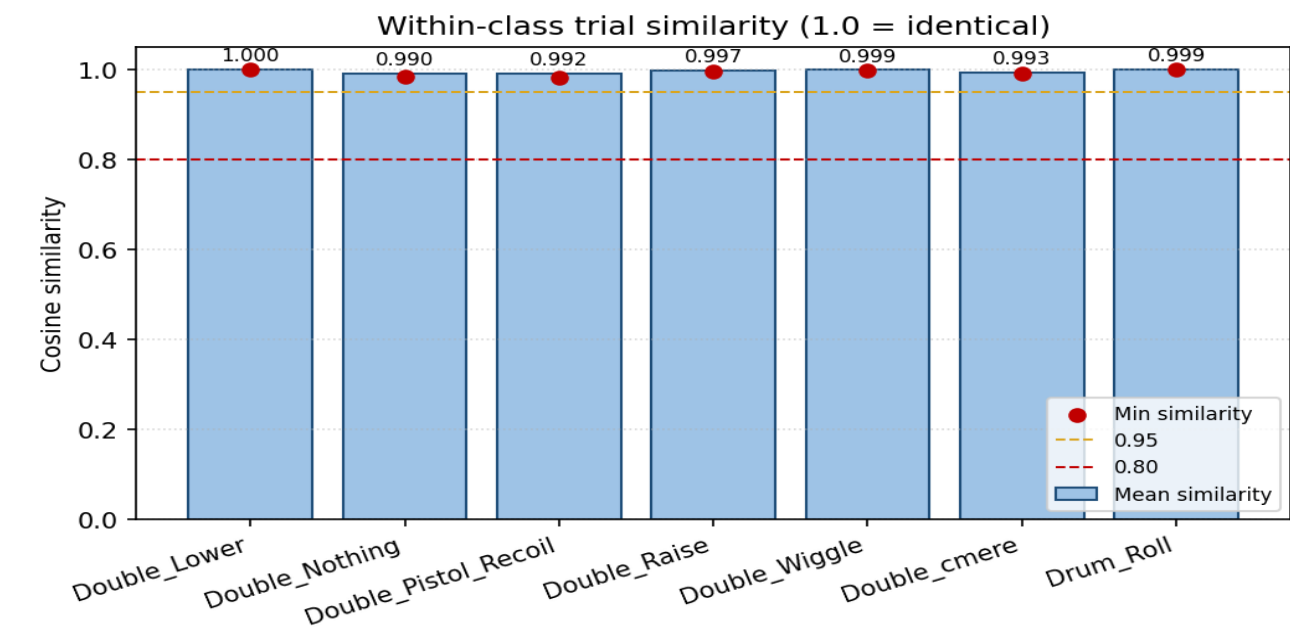


t-SNE feature space projection

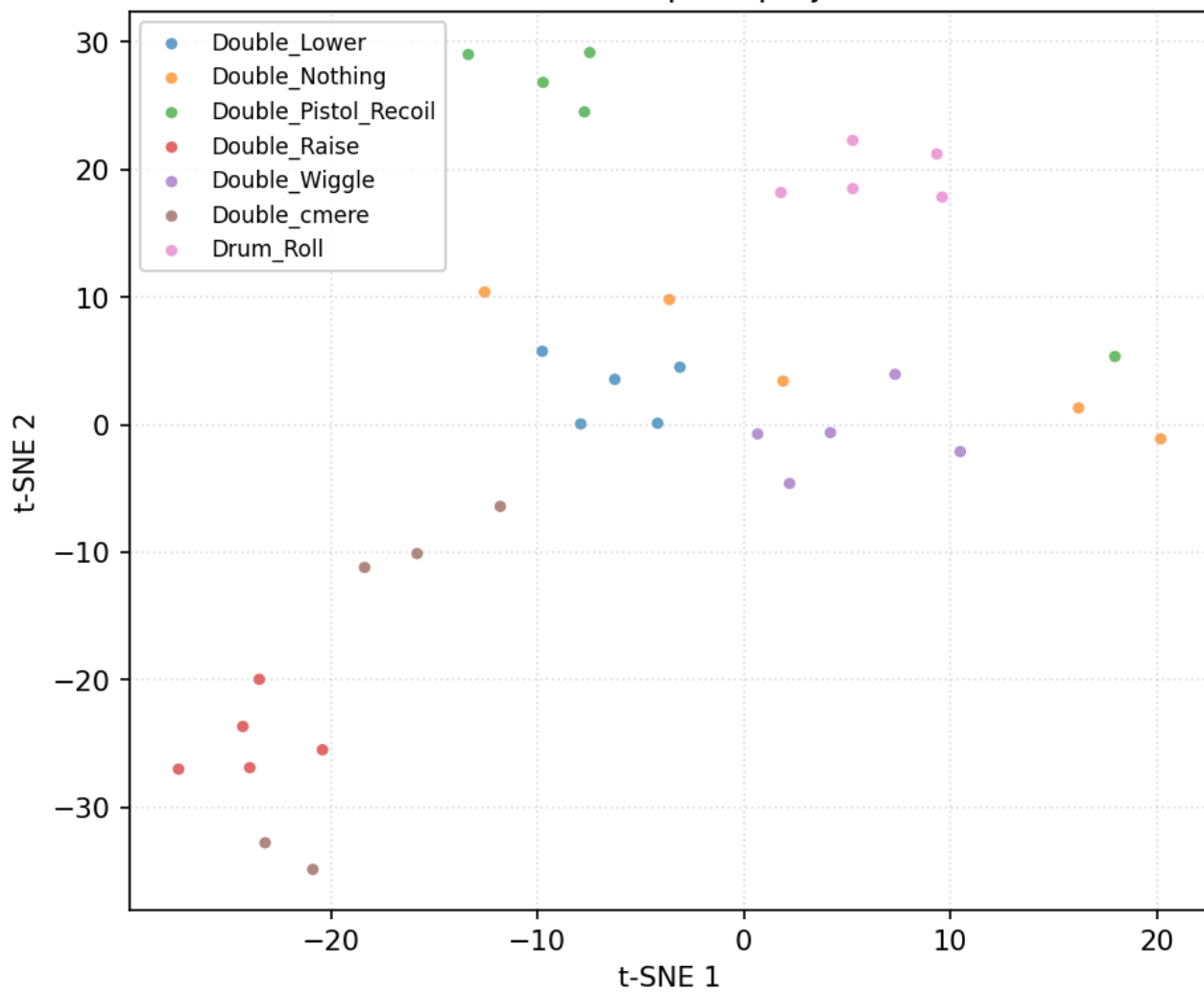


# Data diagnostics — user 15\_Tash

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

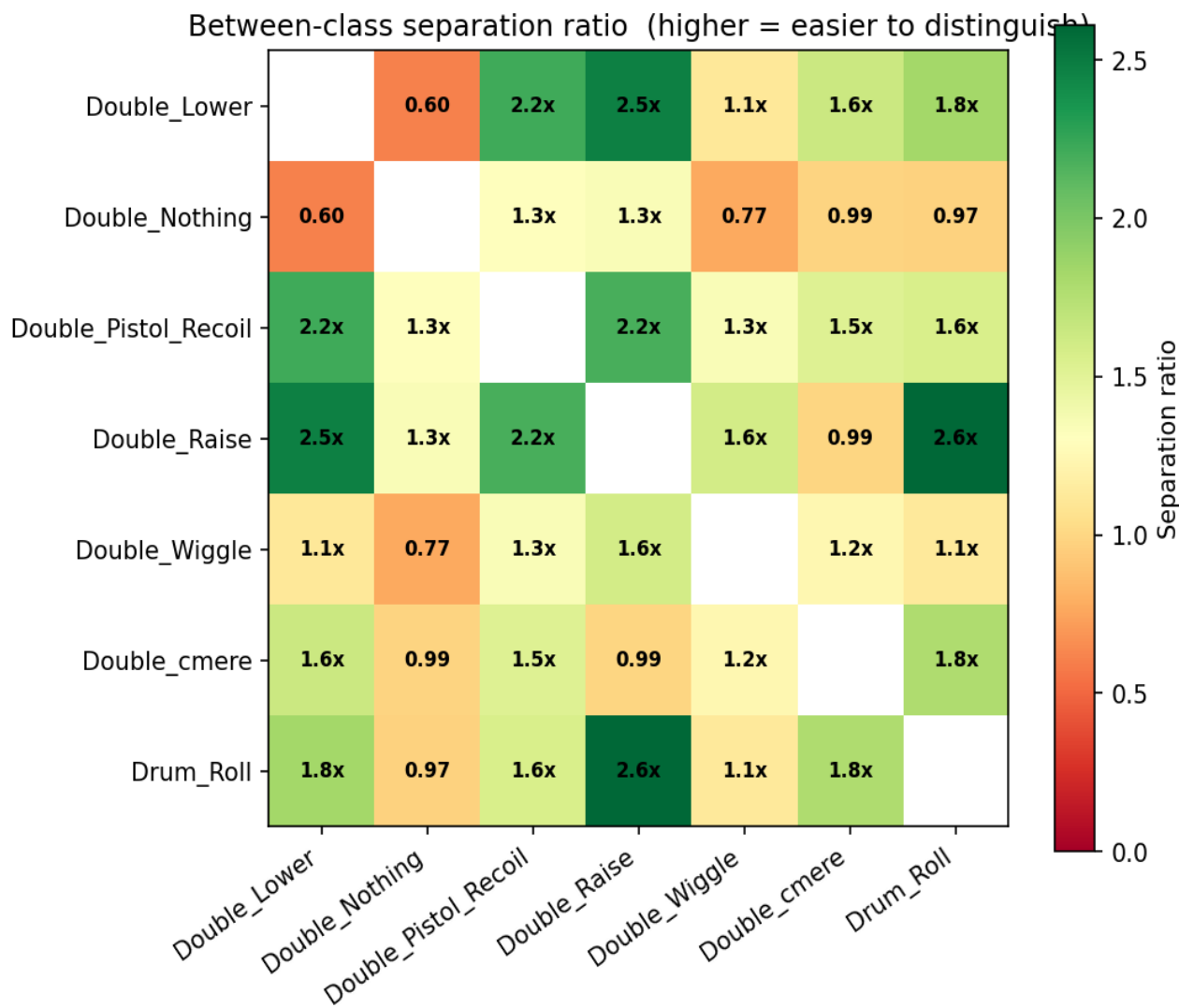
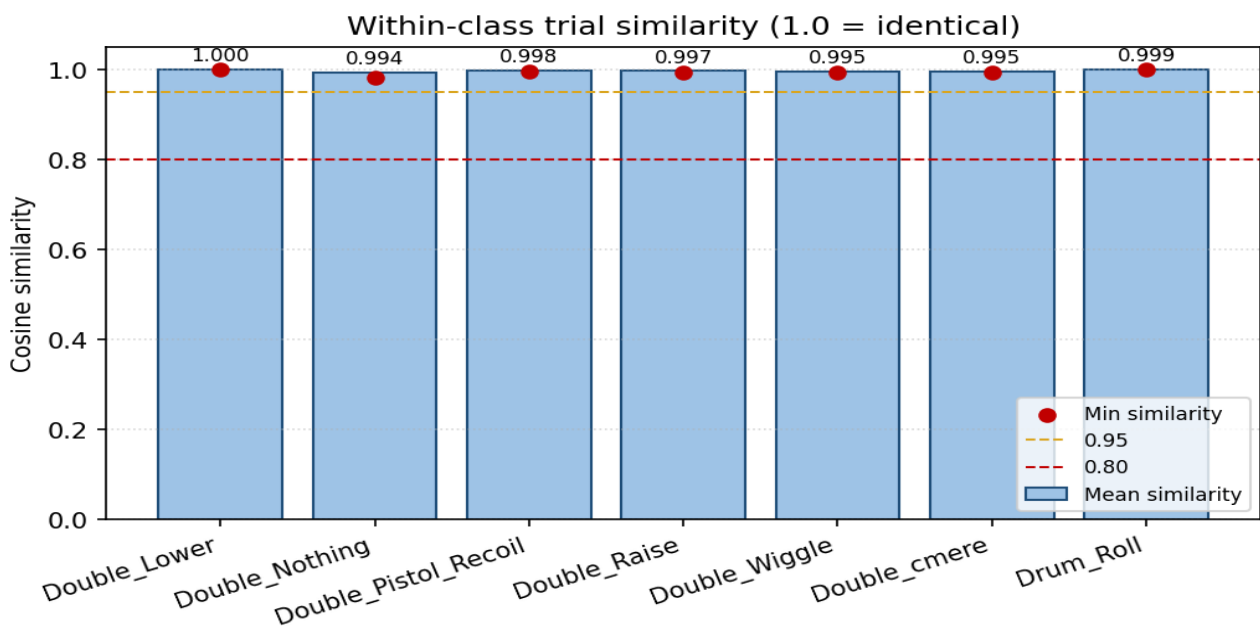


t-SNE feature space projection

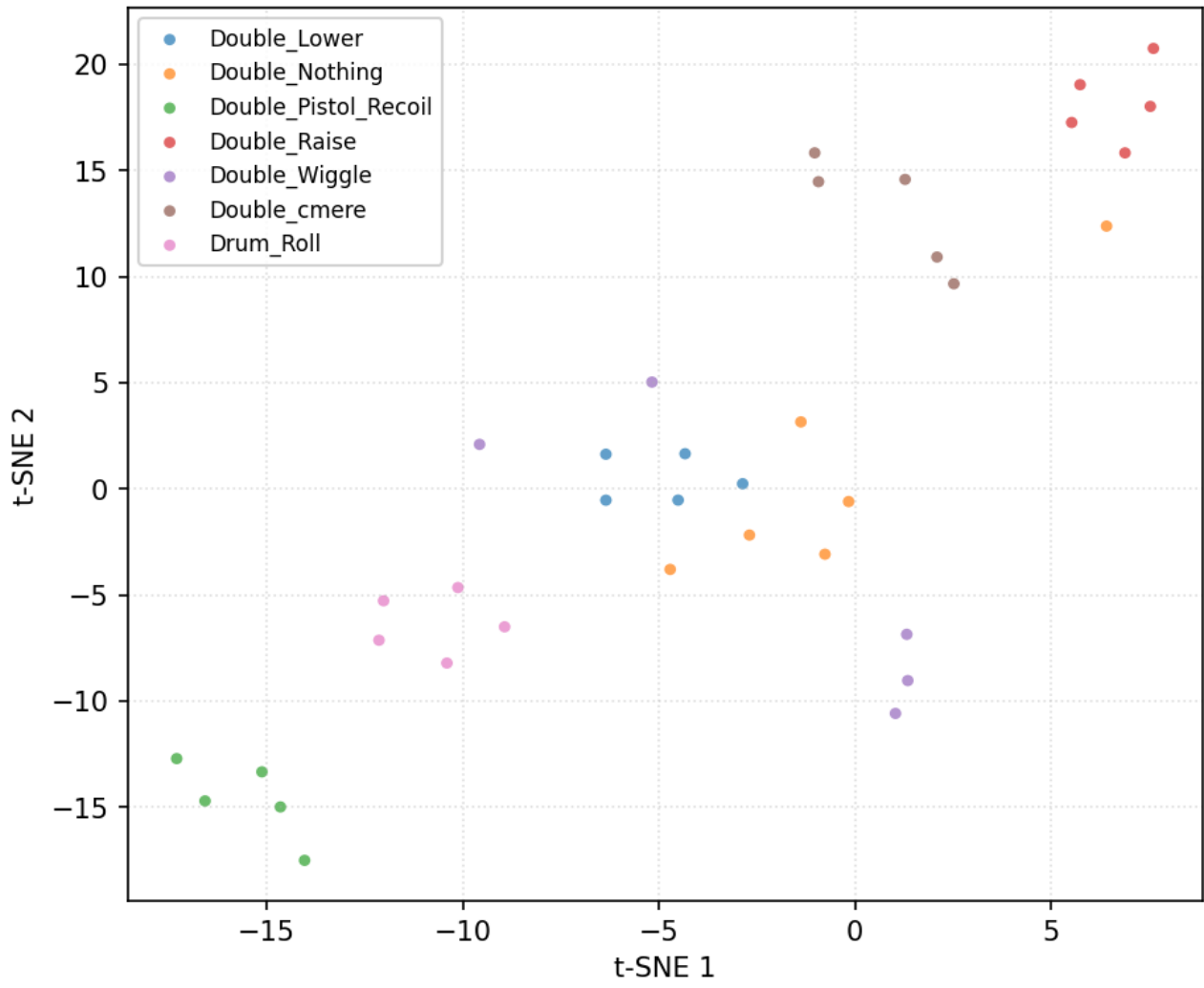


# Data diagnostics — user 16\_Bella

36 trials, 7 classes (filtered + resampled, before per-trial normalisation).

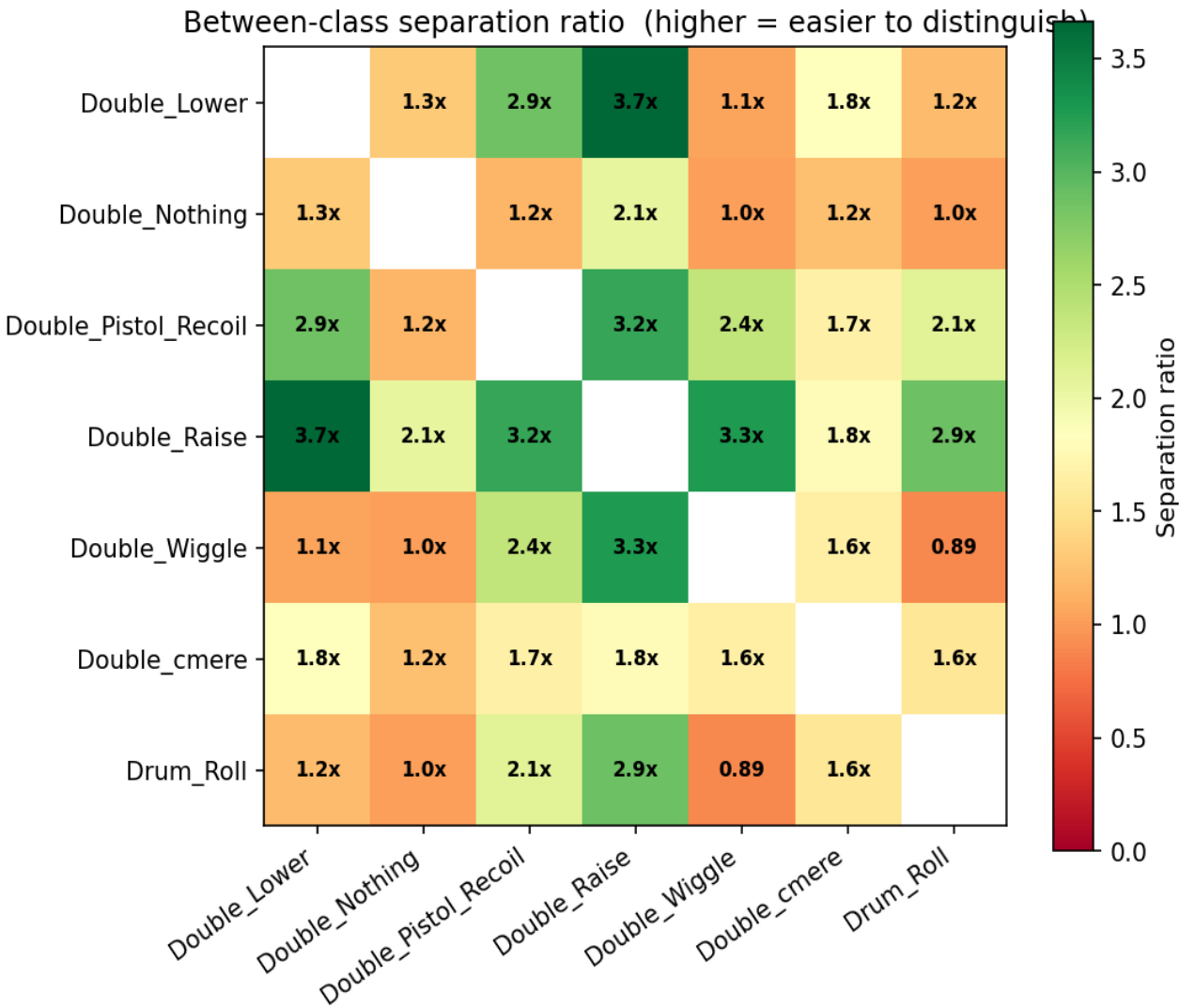
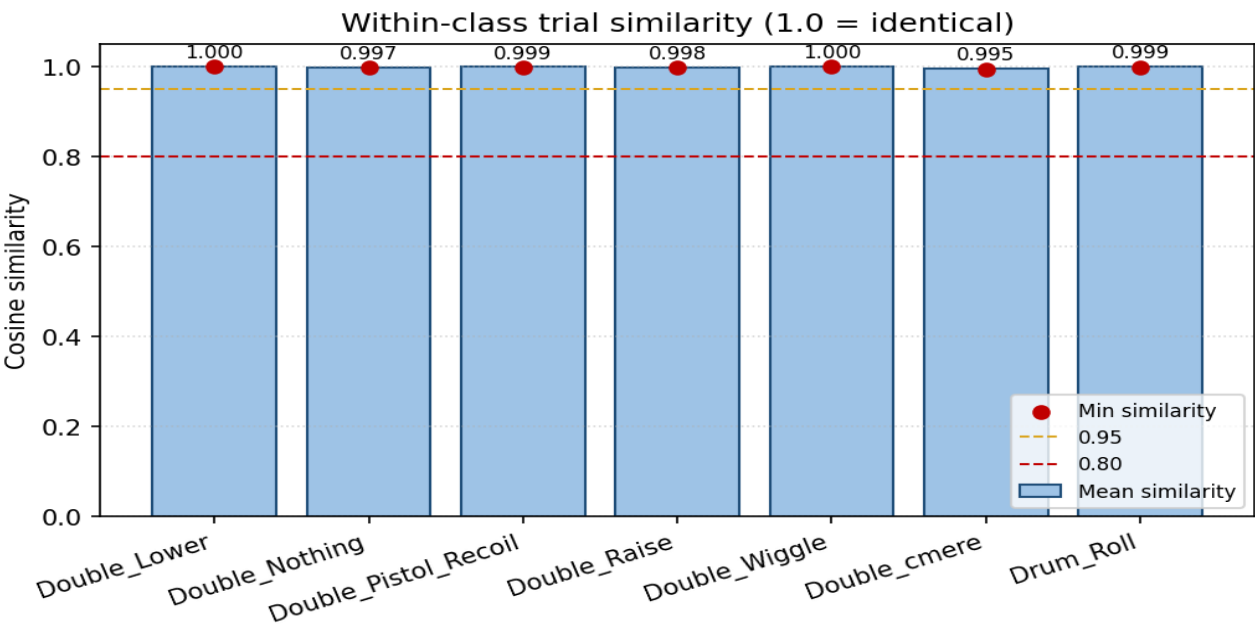


t-SNE feature space projection

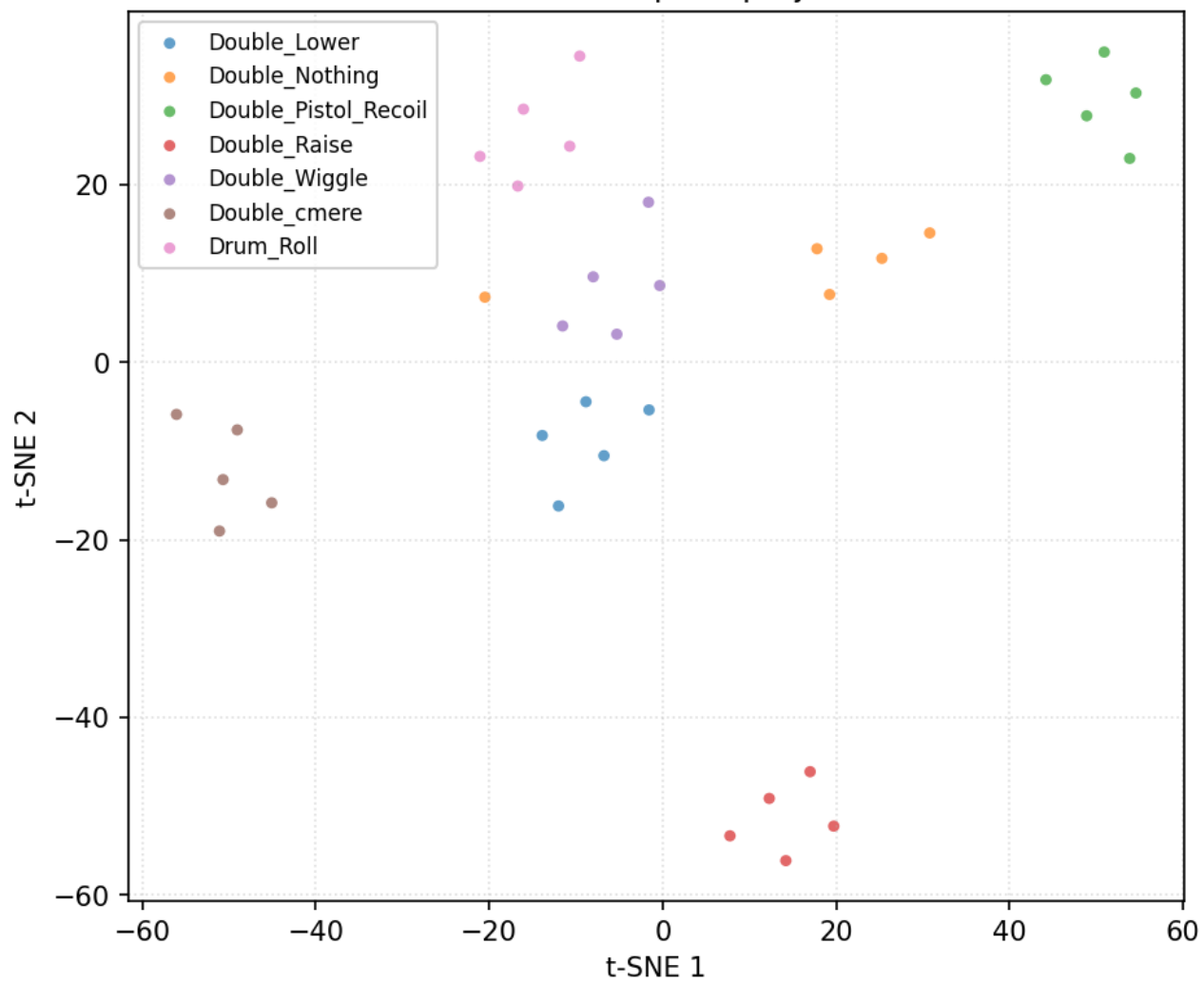


# Data diagnostics — user 17\_Raquel

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).



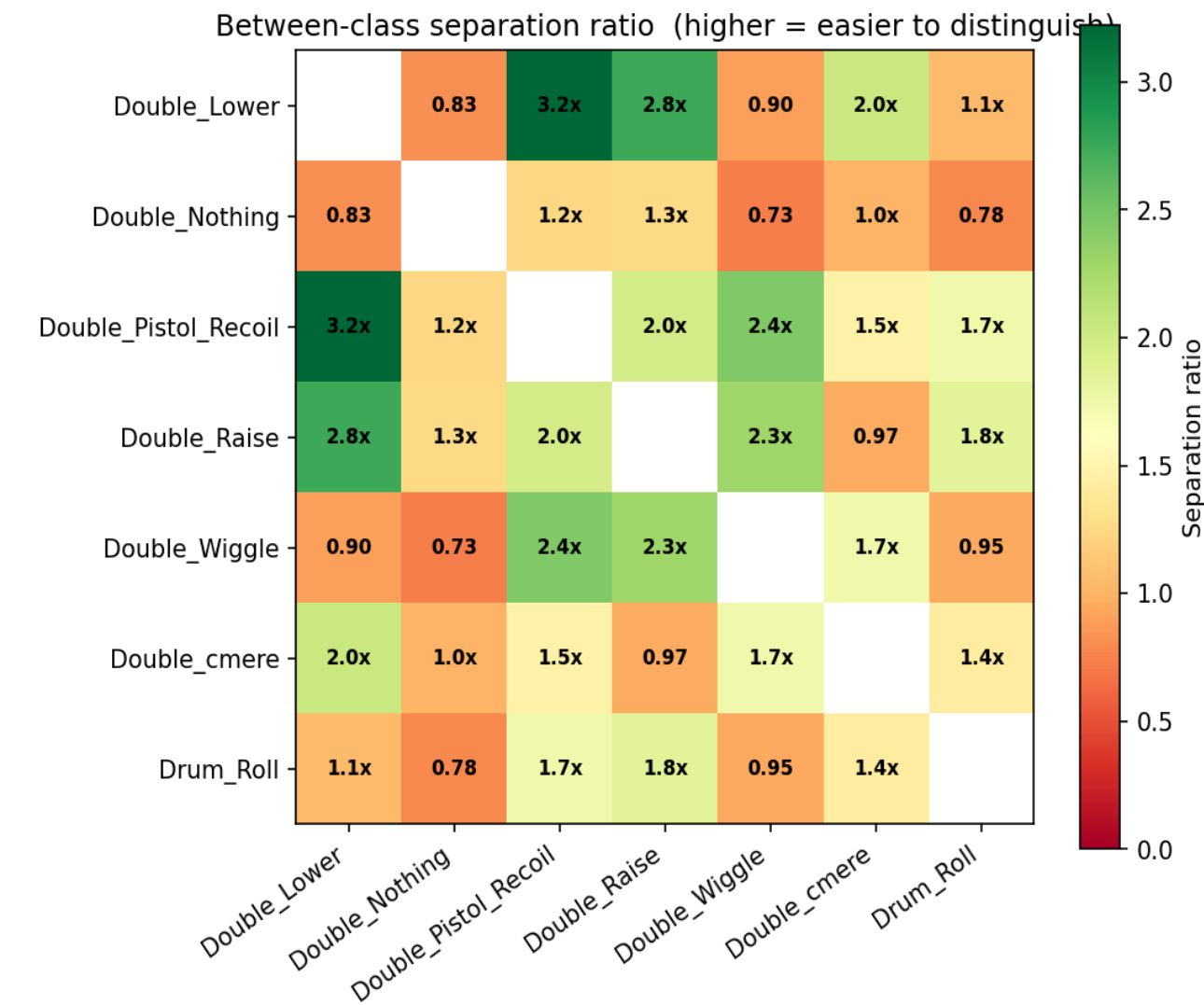
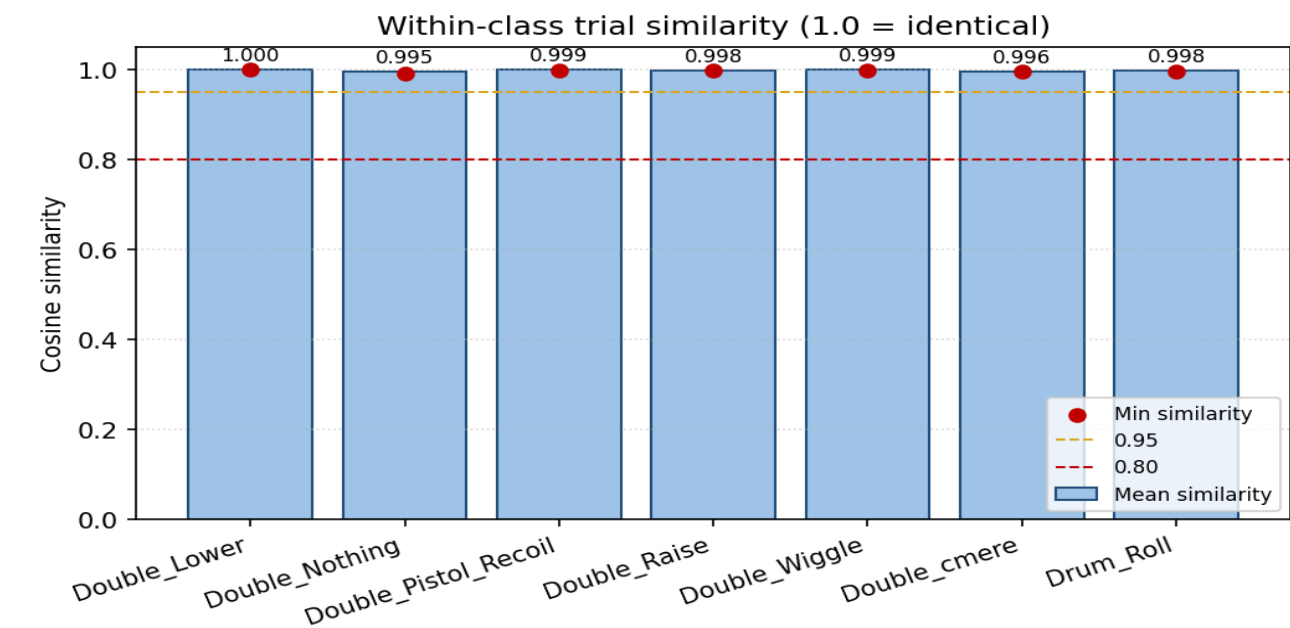
t-SNE feature space projection



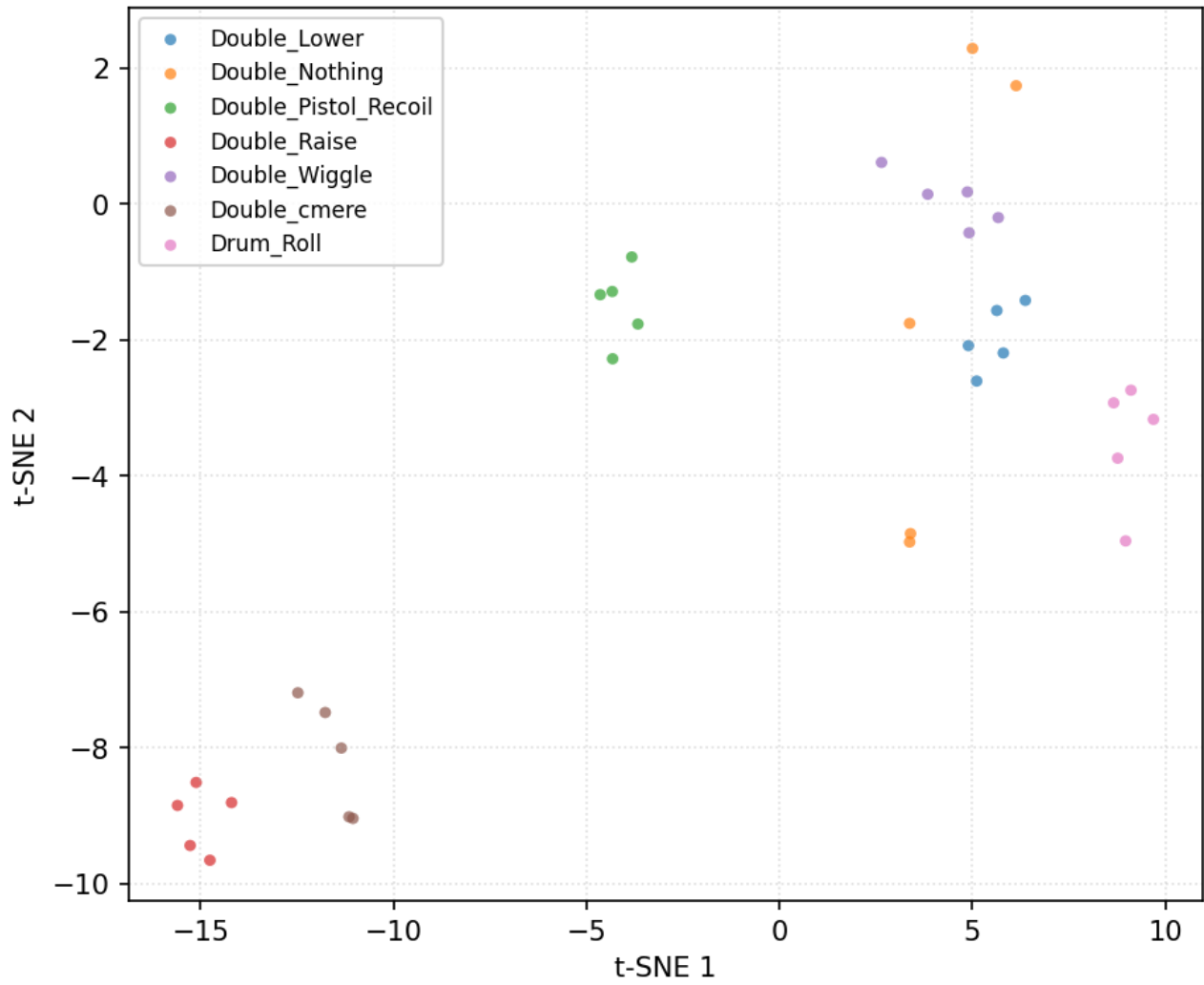


# Data diagnostics — user 18\_Bridgette

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

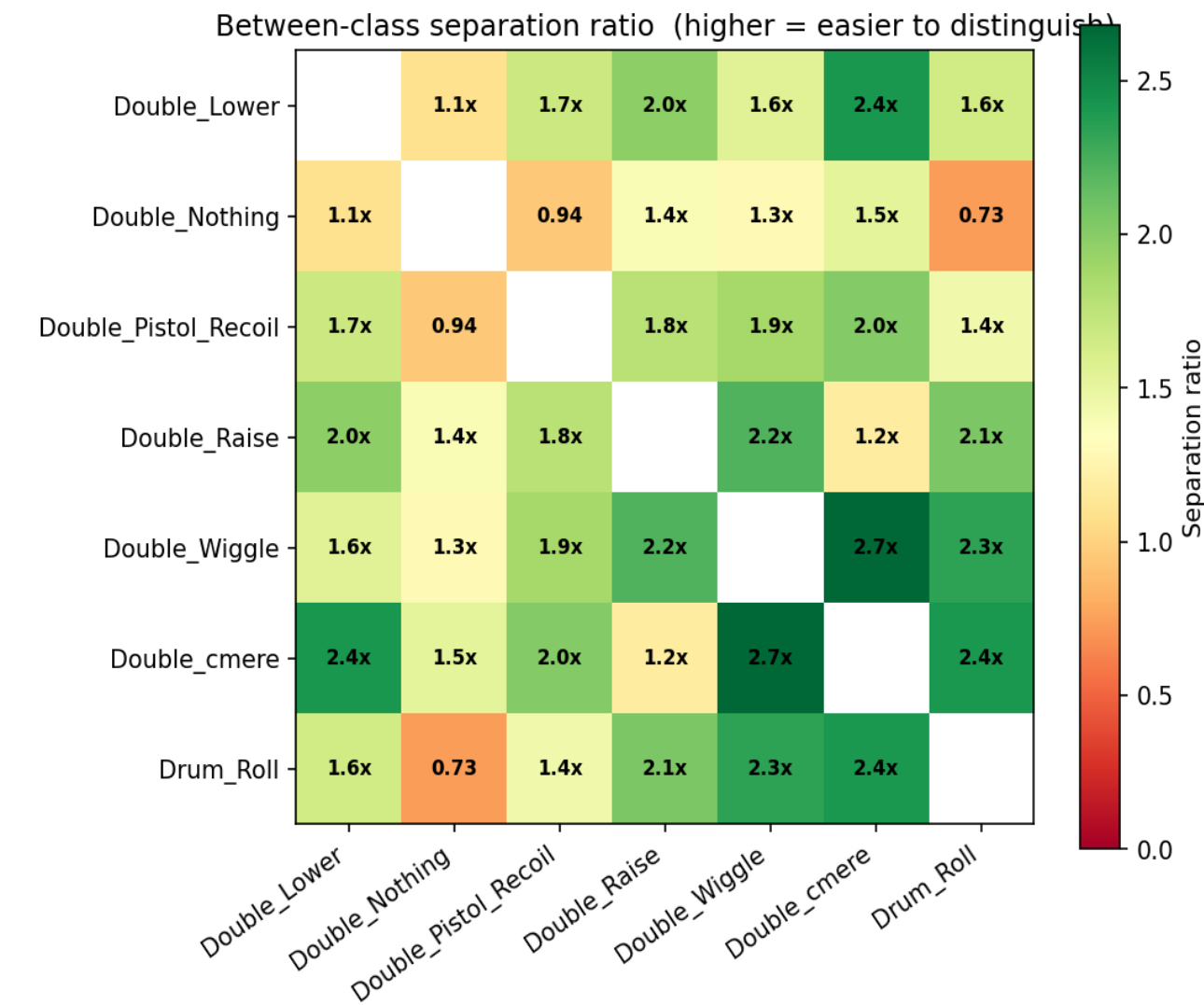
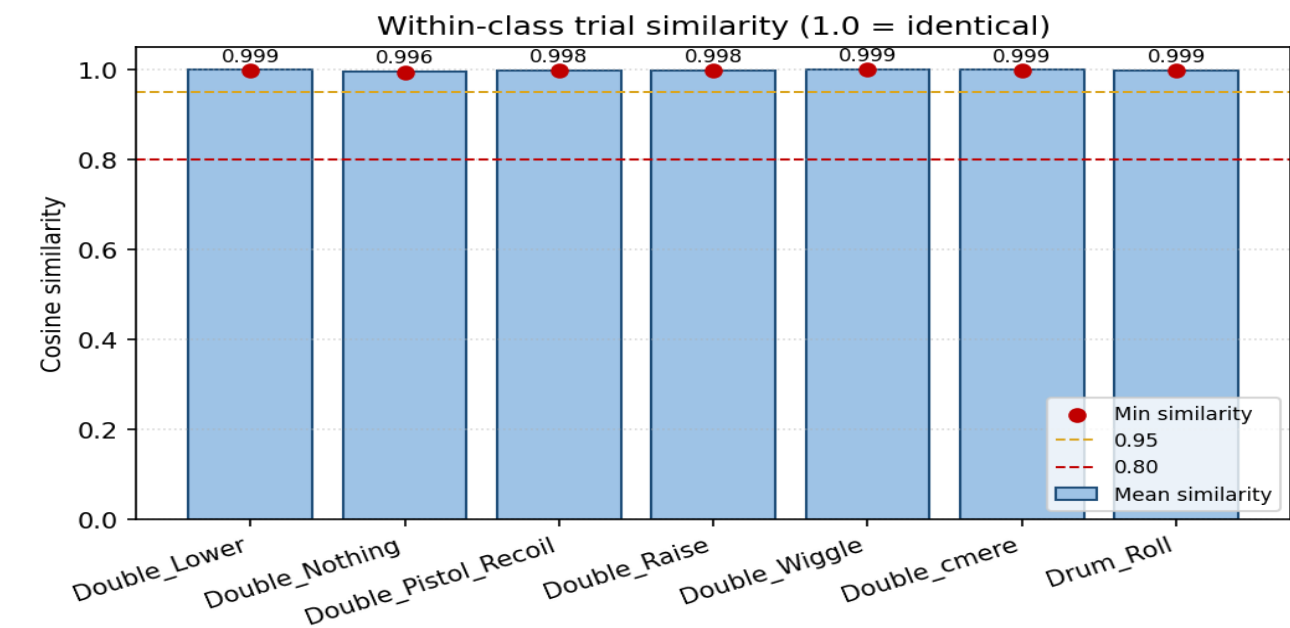


t-SNE feature space projection

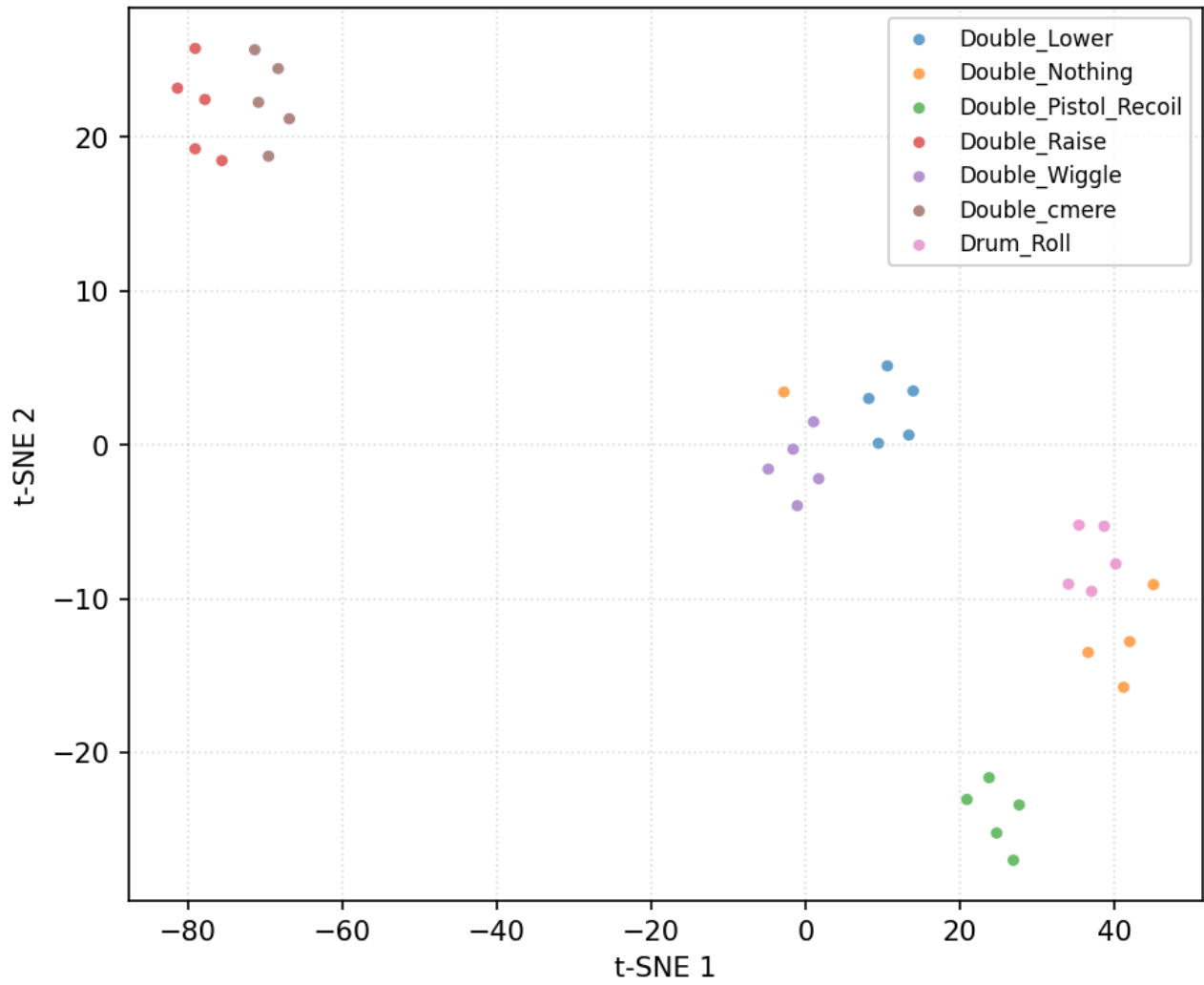


# Data diagnostics — user 1\_Alán

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

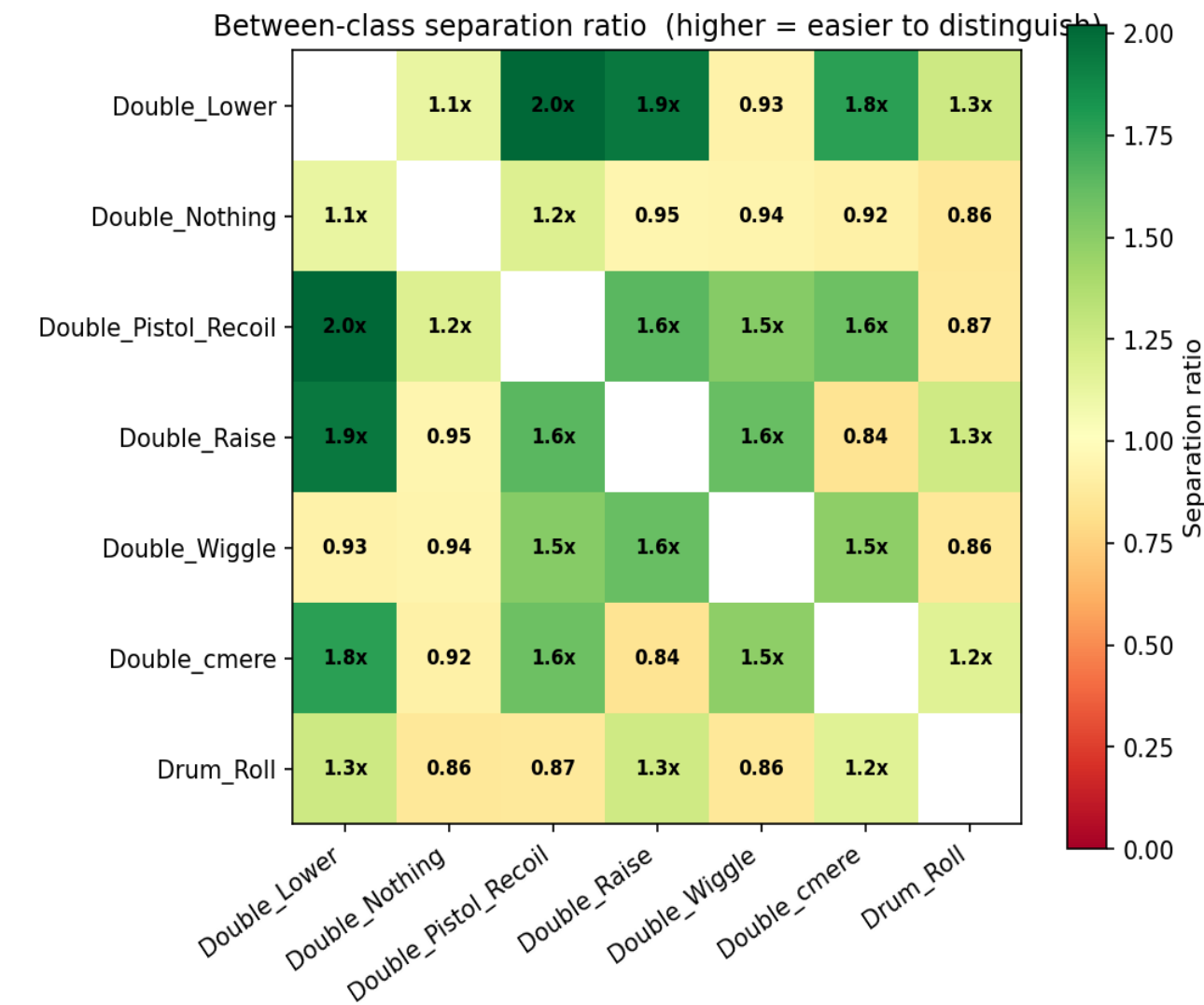
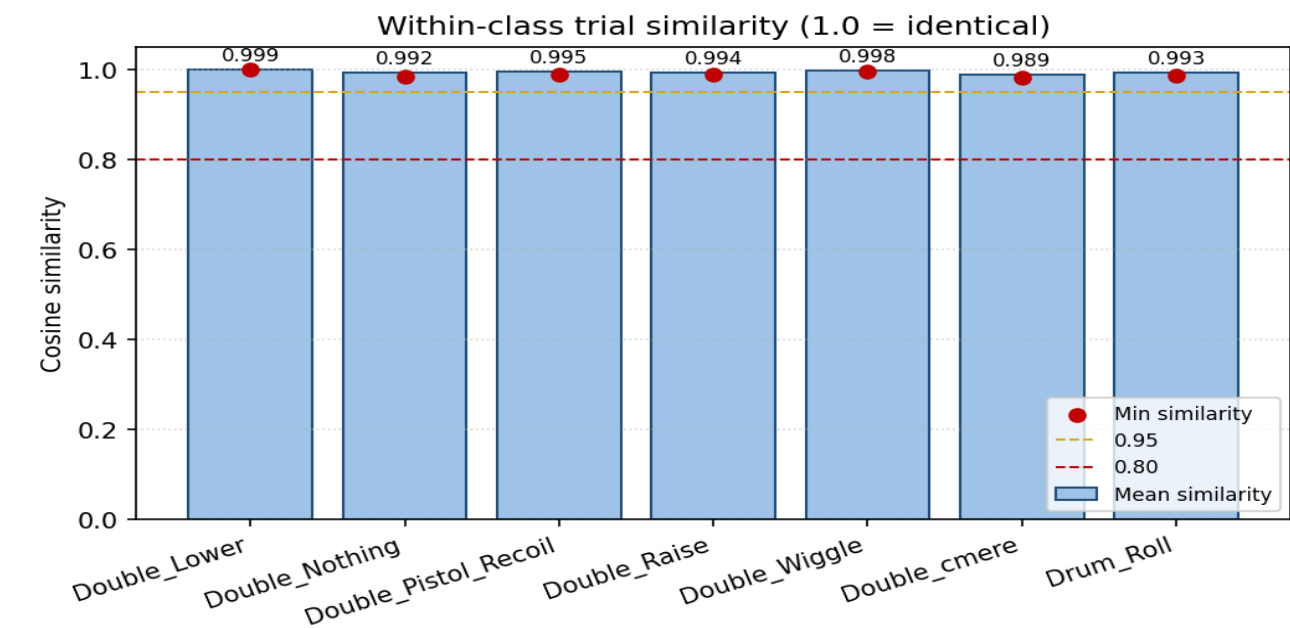


t-SNE feature space projection

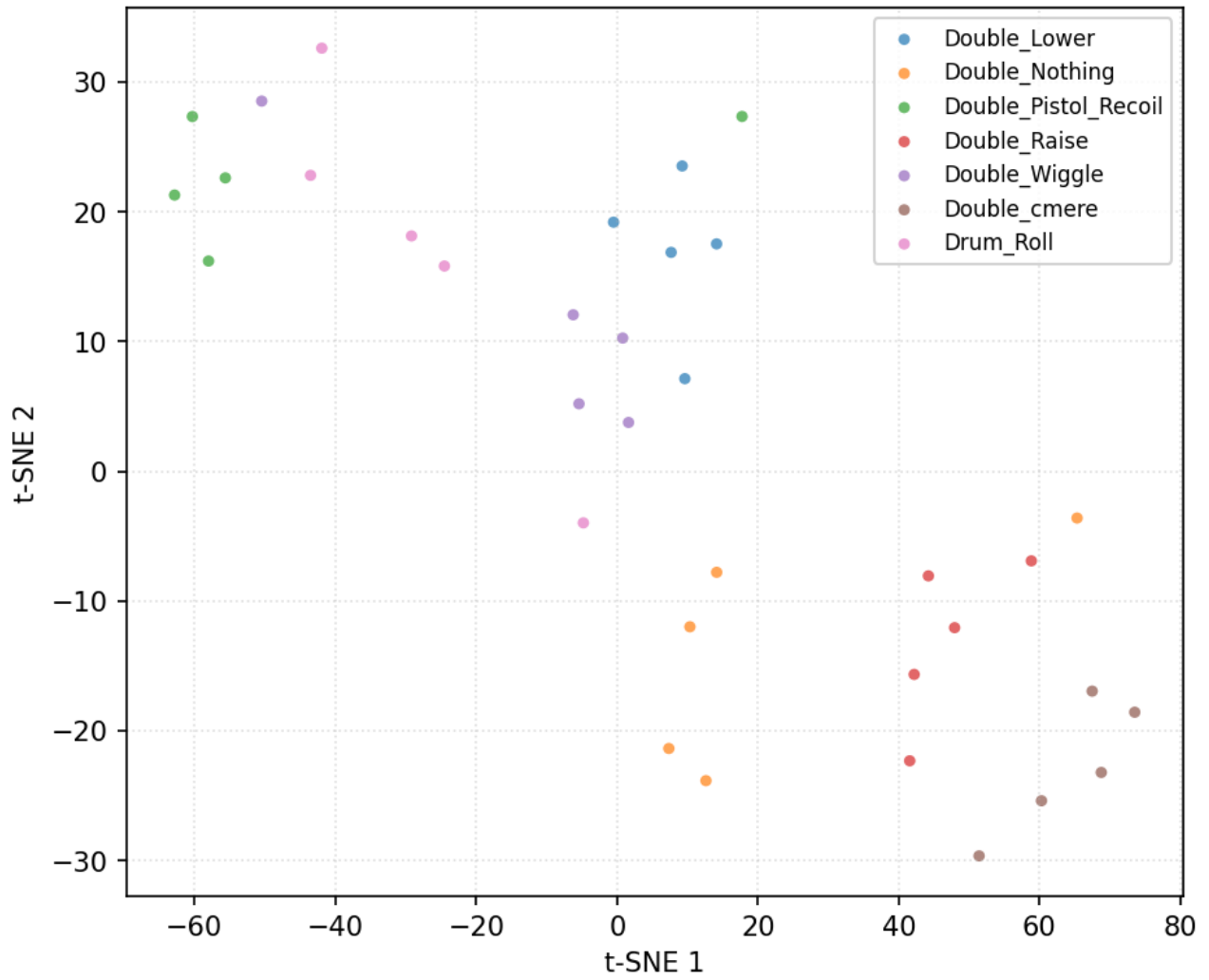


# Data diagnostics — user 2\_Alex

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

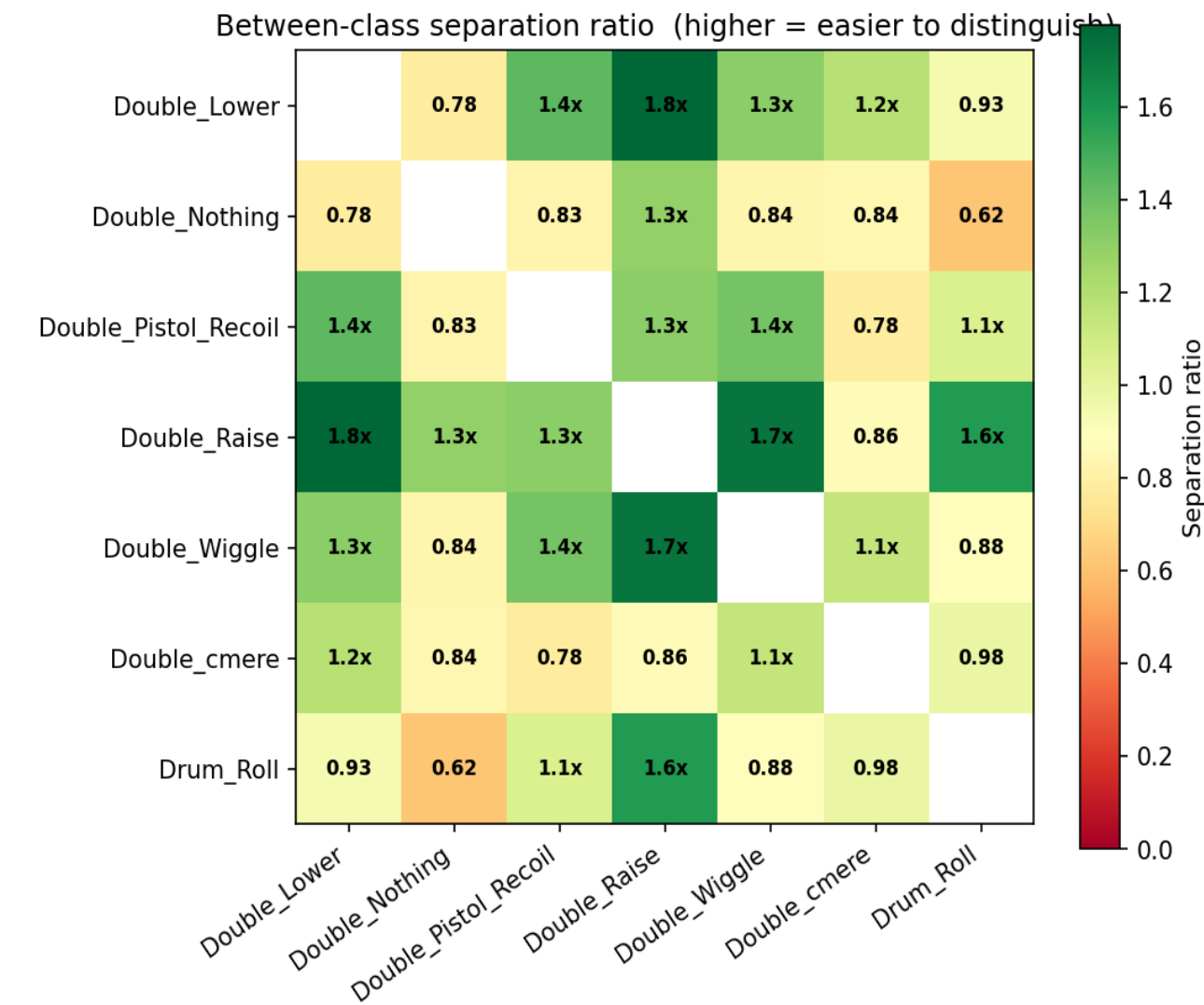
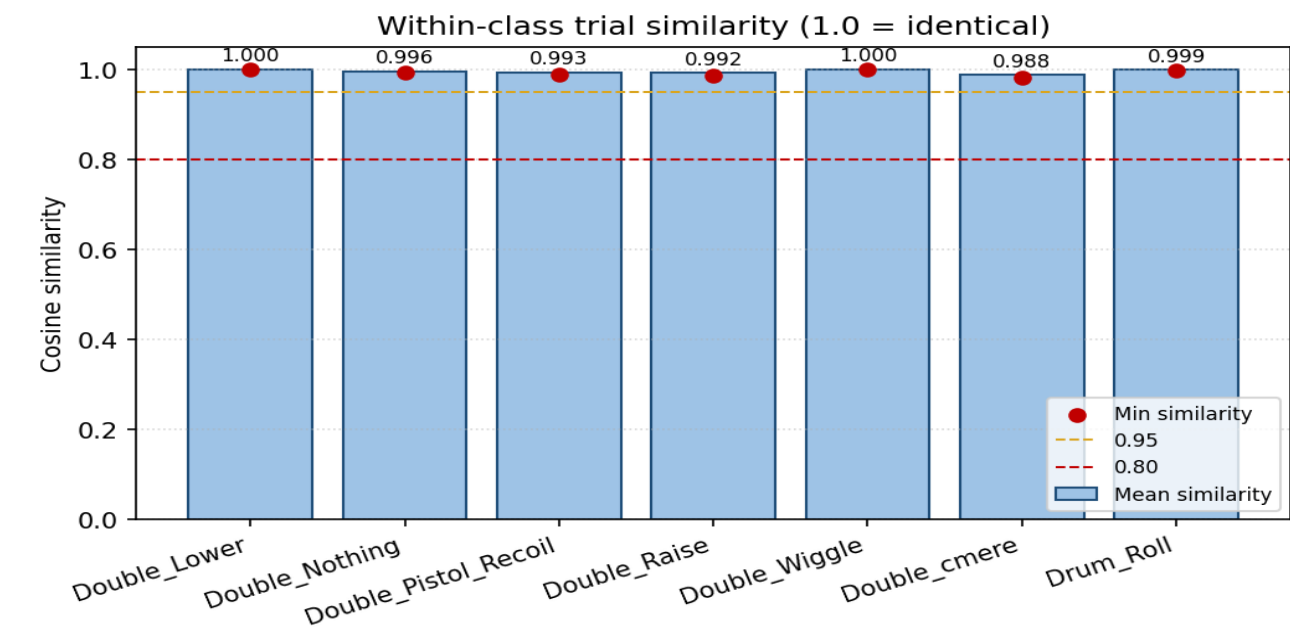


t-SNE feature space projection

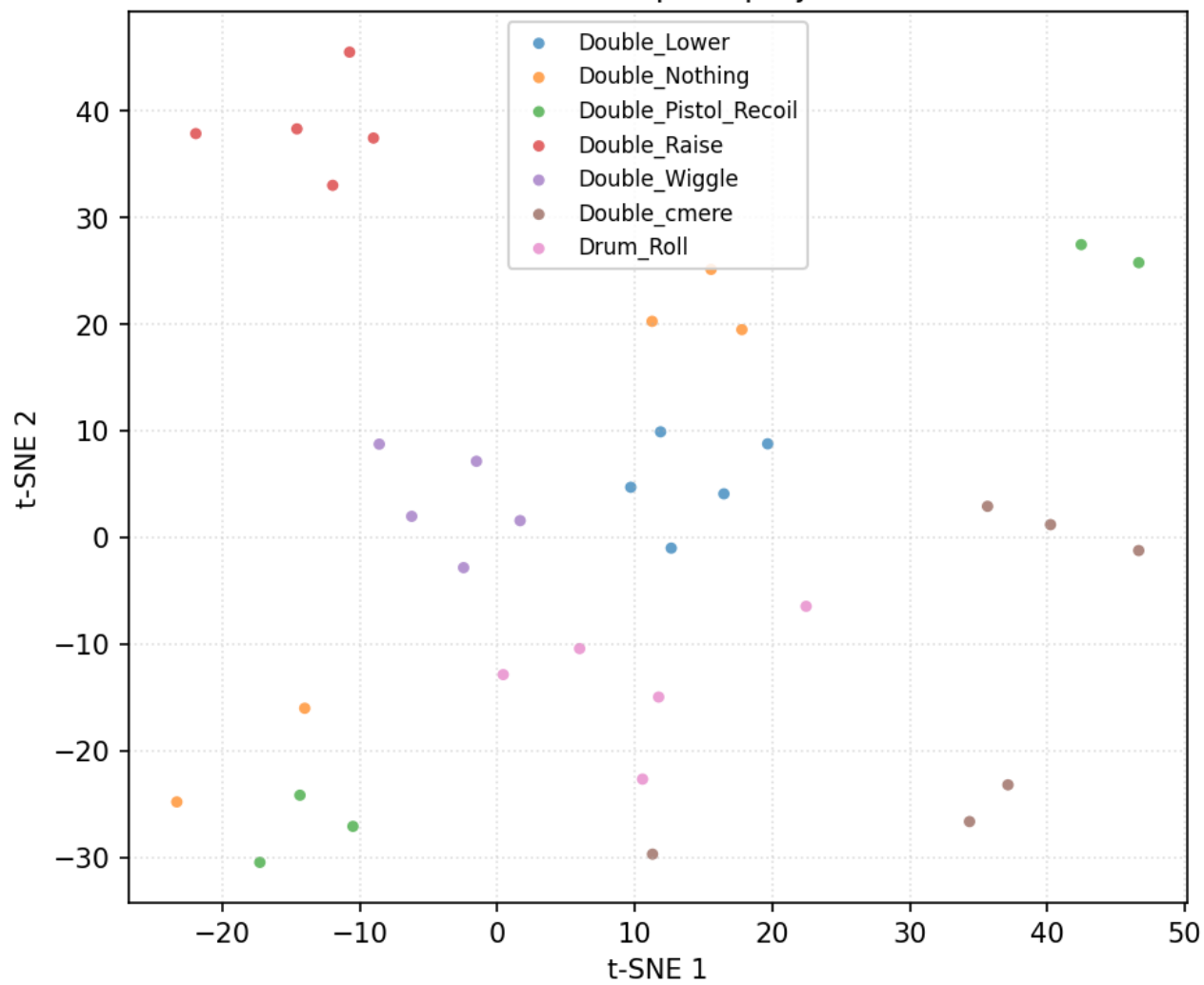


# Data diagnostics — user 3\_Anghad

36 trials, 7 classes (filtered + resampled, before per-trial normalisation).



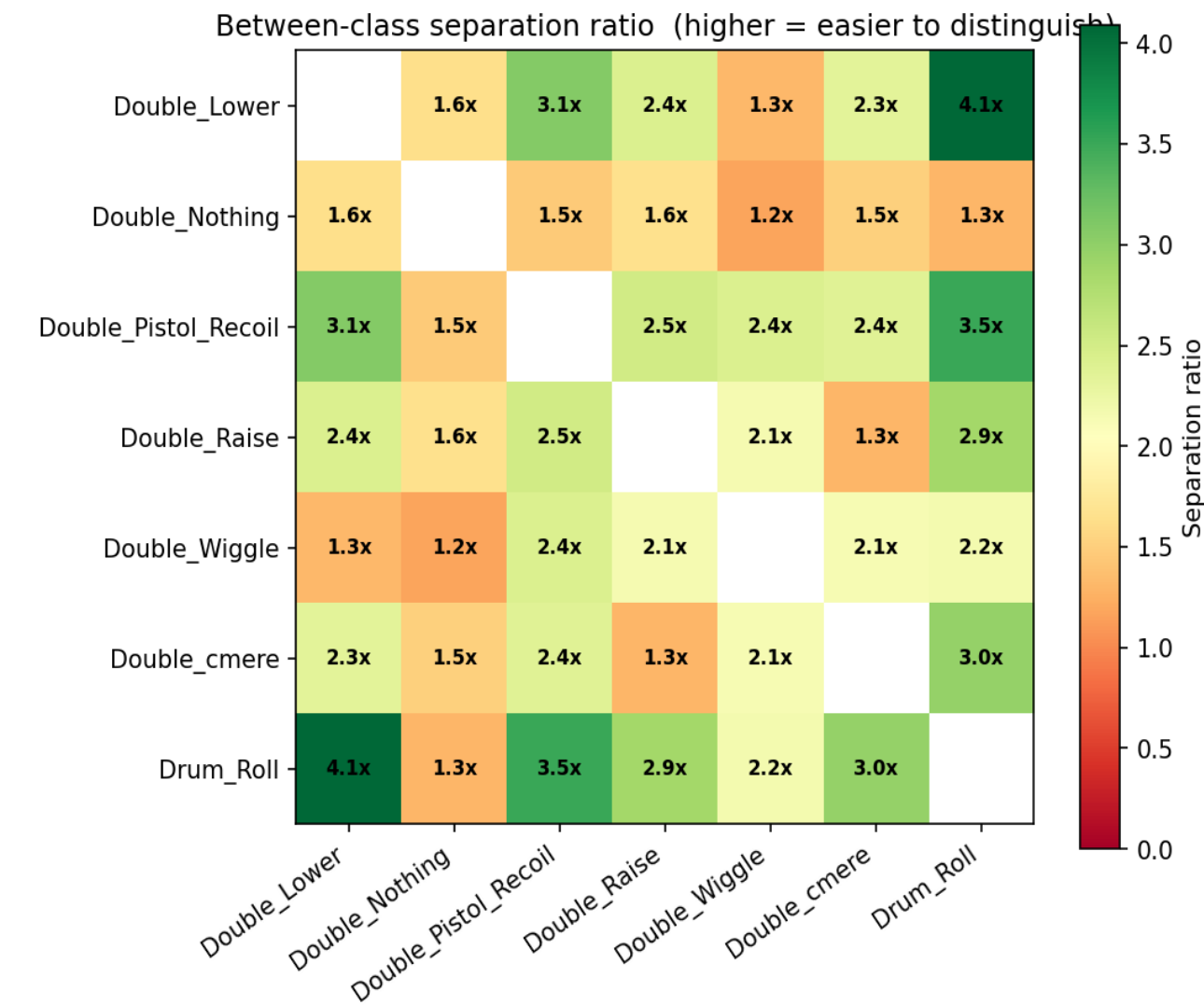
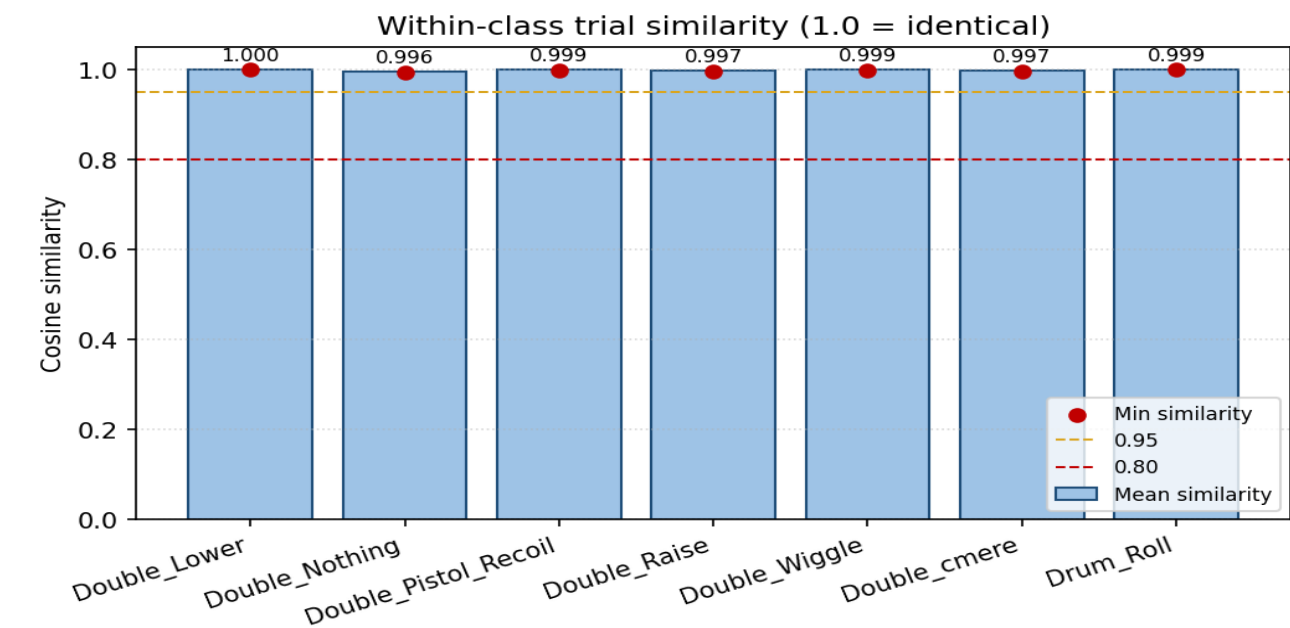
t-SNE feature space projection



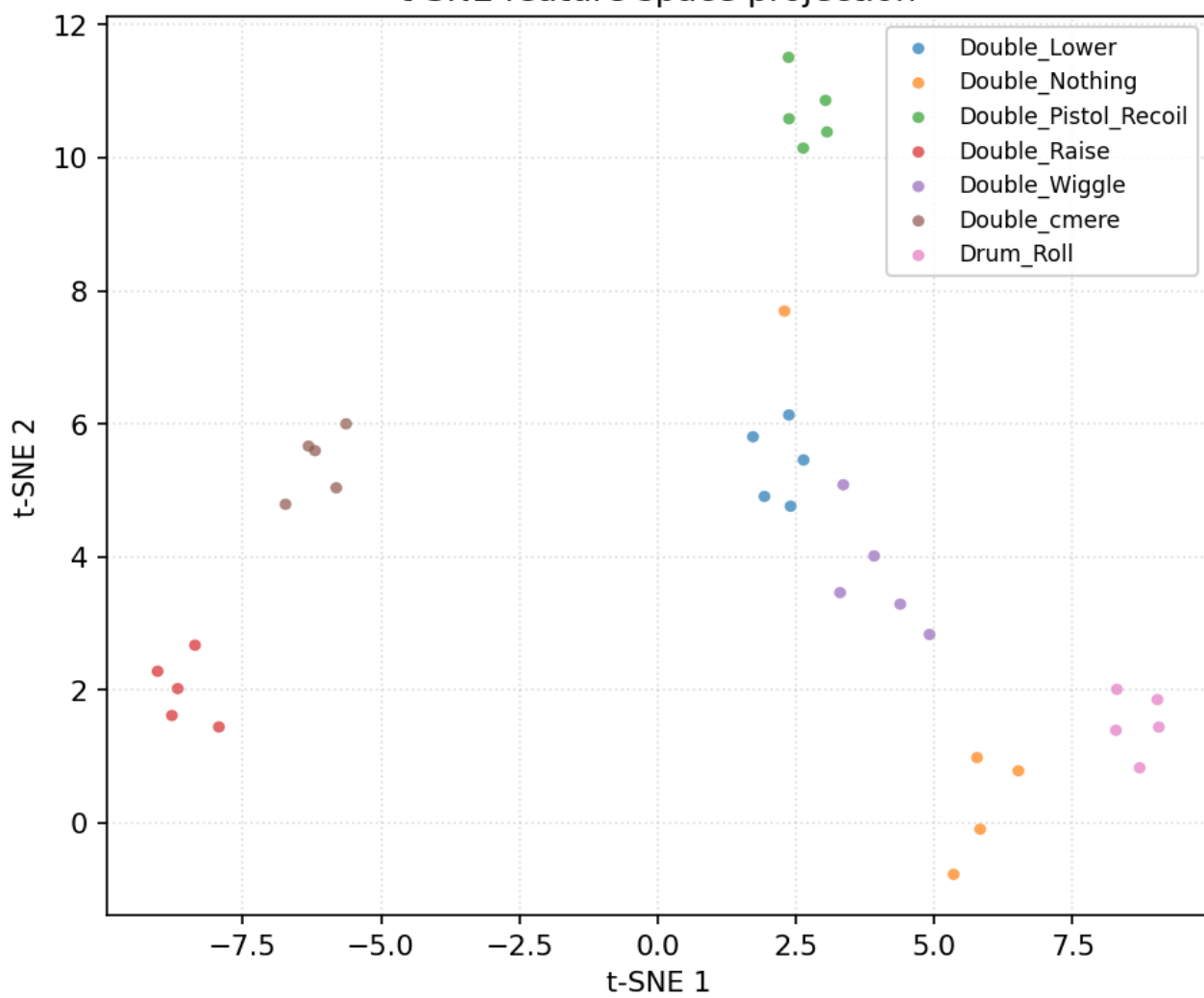


# Data diagnostics — user 4\_Daniel

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

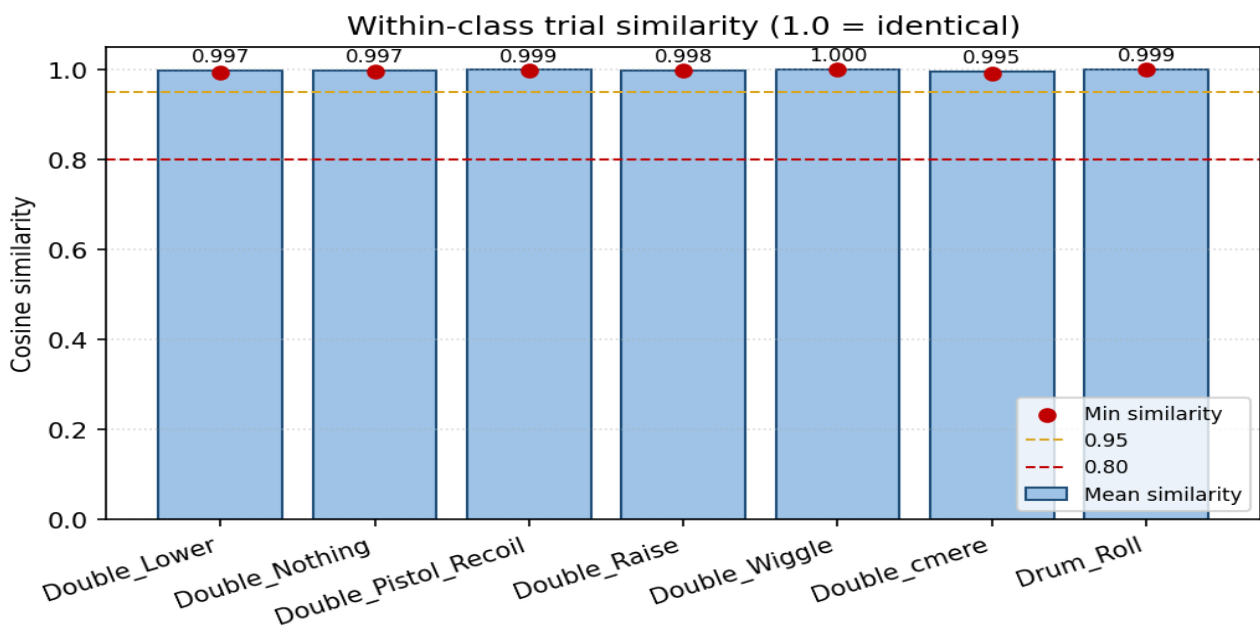


t-SNE feature space projection

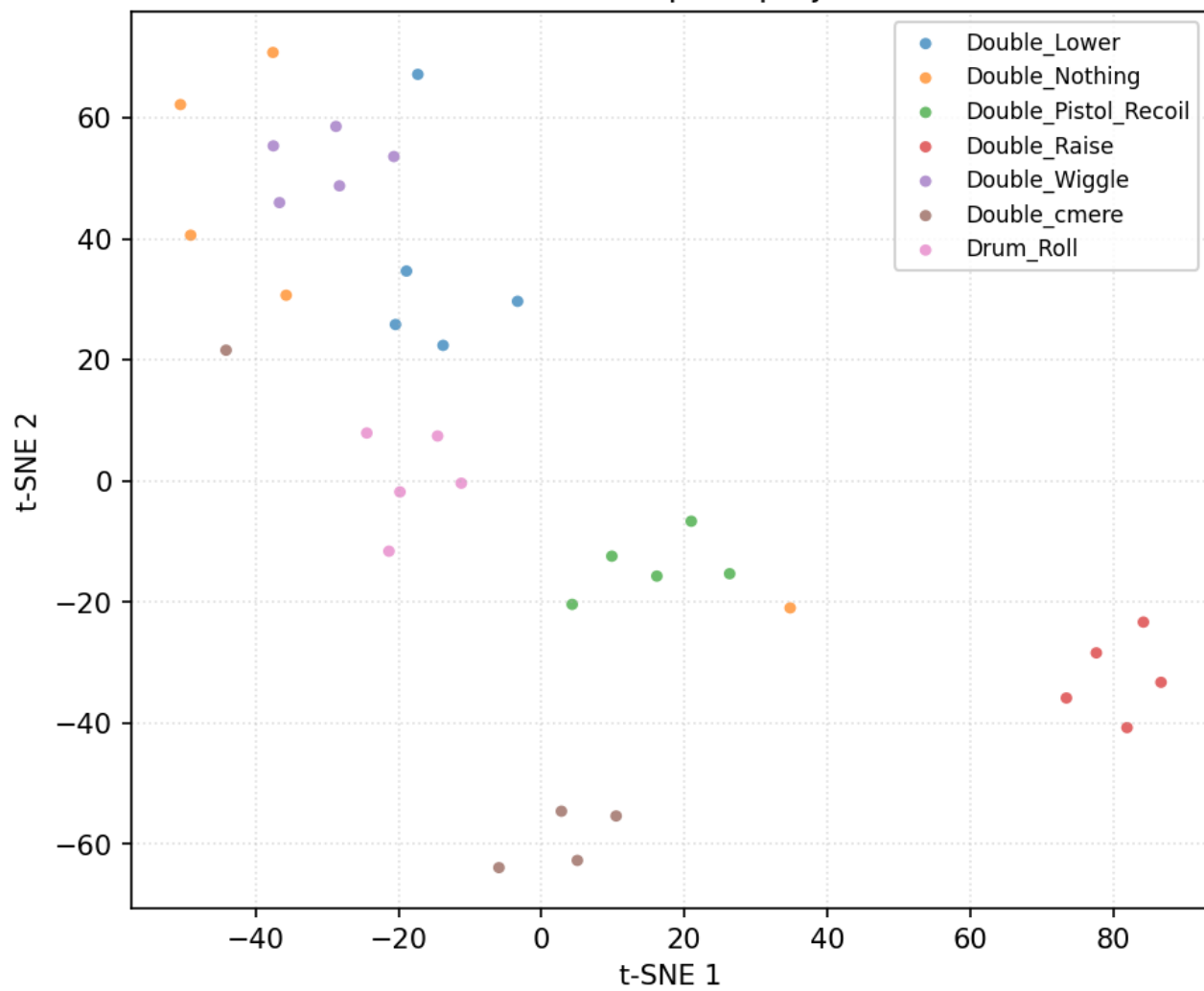


# Data diagnostics — user 6\_Harry

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

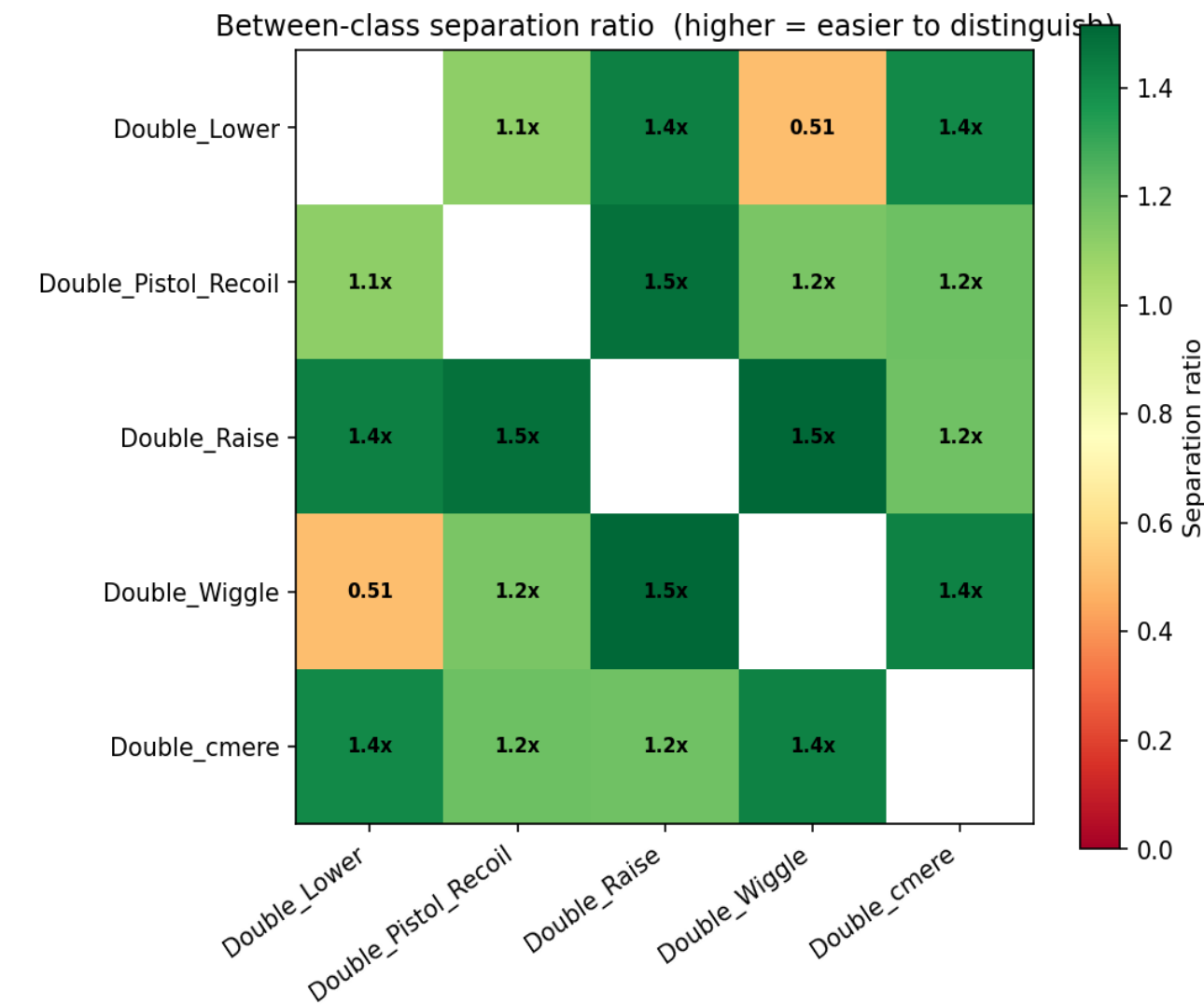
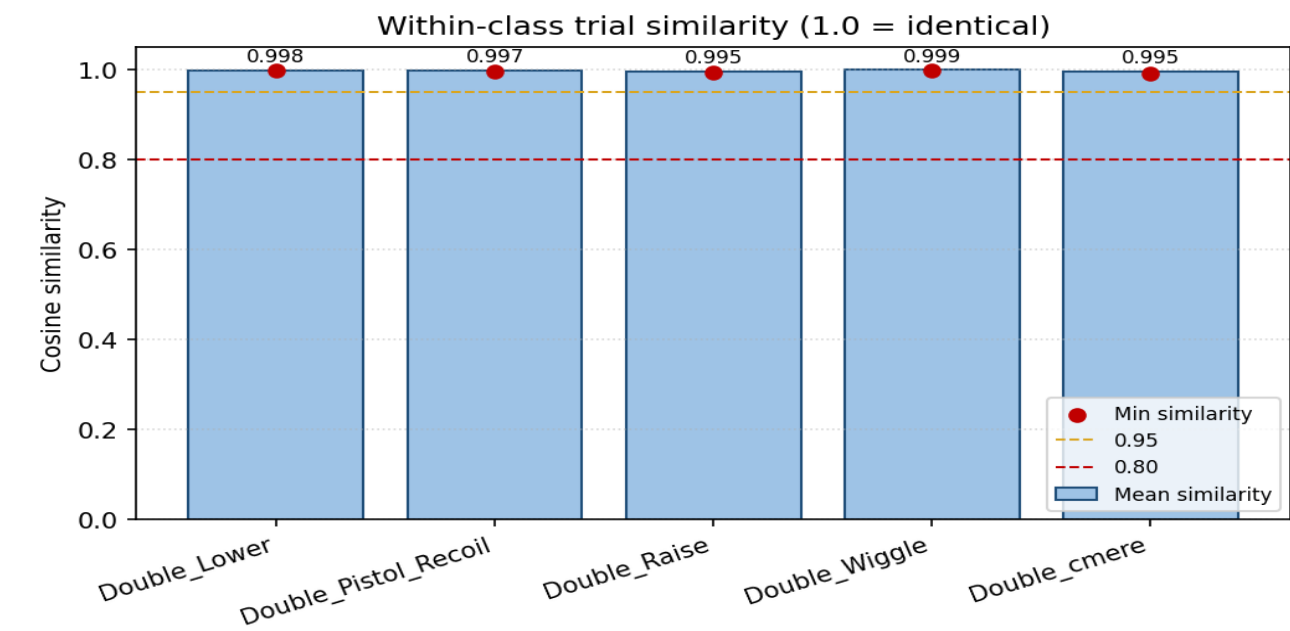


t-SNE feature space projection

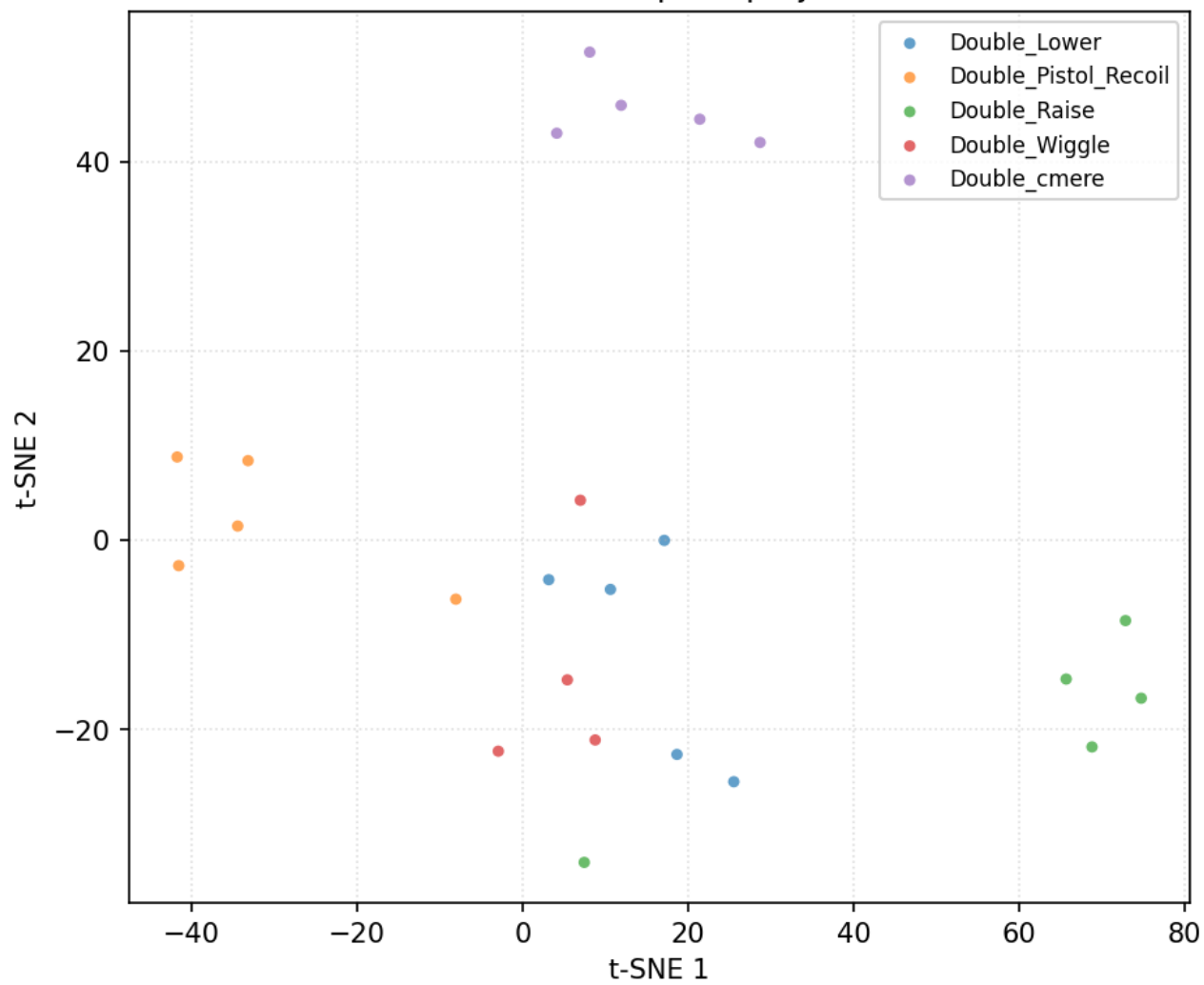


# Data diagnostics — user 8\_Jestin

24 trials, 5 classes (filtered + resampled, before per-trial normalisation).



t-SNE feature space projection



## Variant: BN\_GAP

Conv+BN+ReLU x3 with GlobalAveragePooling1D head — fewer params, less overfit

|                                       |                     |
|---------------------------------------|---------------------|
| <b>Trainable params</b>               | 57,319              |
| <b>LOUO mean <math>\pm</math> std</b> | 0.9294 $\pm$ 0.0663 |
| <b>Folds run</b>                      | 13                  |
| <b>Inner-val mode</b>                 | user                |
| <b>Per-trial norm</b>                 | zscore              |

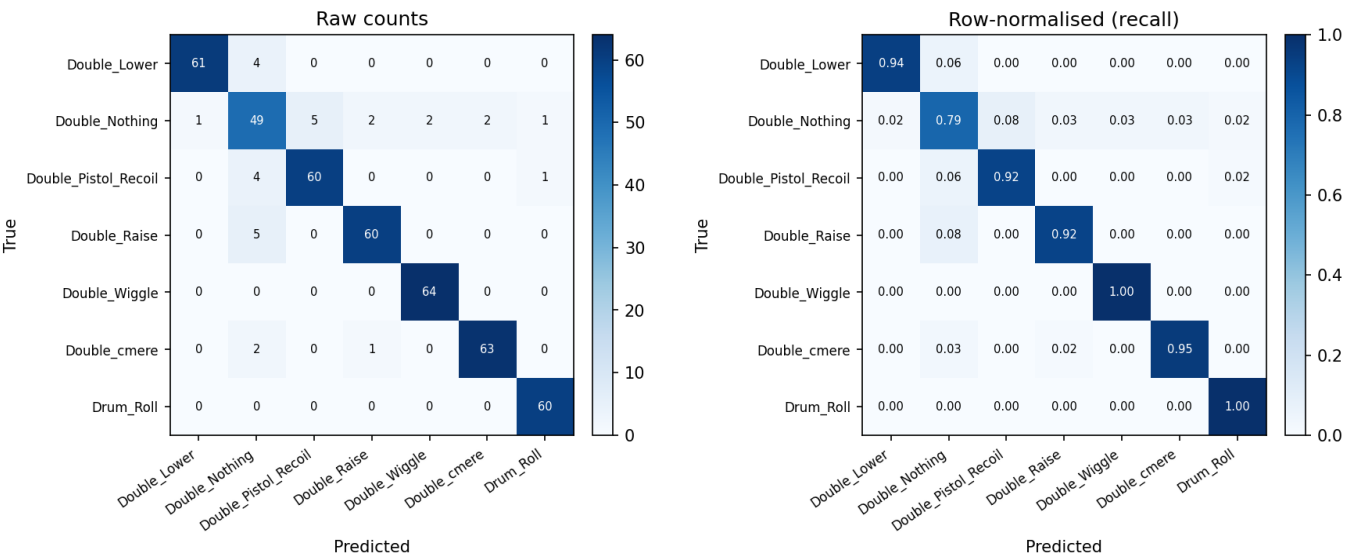
### LOUO per-fold results

| Fold | Held-out user | Inner-val user | n_train | n_test | Accuracy |
|------|---------------|----------------|---------|--------|----------|
| 1    | 11_Mansh      | 12_Marcus      | 377     | 35     | 0.8571   |
| 2    | 12_Marcus     | 14_StephenV2   | 376     | 35     | 1.0000   |
| 3    | 14_StephenV2  | 15_Tash        | 376     | 36     | 0.9444   |
| 4    | 15_Tash       | 16_Bella       | 376     | 35     | 0.9714   |
| 5    | 16_Bella      | 17_Raquel      | 376     | 36     | 0.9167   |
| 6    | 17_Raquel     | 18_Bridgette   | 377     | 35     | 1.0000   |
| 7    | 18_Bridgette  | 1_Alan         | 377     | 35     | 0.9714   |
| 8    | 1_Alan        | 2_Alex         | 377     | 35     | 0.8286   |
| 9    | 2_Alex        | 3_Anghad       | 376     | 35     | 0.9429   |
| 10   | 3_Anghad      | 4_Daniel       | 376     | 36     | 0.9722   |
| 11   | 4_Daniel      | 6_Harry        | 377     | 35     | 1.0000   |
| 12   | 6_Harry       | 8_Jestin       | 388     | 35     | 0.8857   |
| 13   | 8_Jestin      | 11_Mansh       | 388     | 24     | 0.7917   |

### Per-class metrics (aggregated across folds)

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 0.984     | 0.938  | 0.961 | 65      |
| Double_Nothing       | 0.766     | 0.790  | 0.778 | 62      |
| Double_Pistol_Recoil | 0.923     | 0.923  | 0.923 | 65      |
| Double_Raise         | 0.952     | 0.923  | 0.938 | 65      |
| Double_Wiggle        | 0.970     | 1.000  | 0.985 | 64      |
| Double_cmere         | 0.969     | 0.955  | 0.962 | 66      |
| Drum_Roll            | 0.968     | 1.000  | 0.984 | 60      |
| macro avg            | 0.933     | 0.933  | 0.933 | 447     |
| weighted avg         | 0.934     | 0.933  | 0.933 | 447     |

### Confusion matrix (aggregated across folds)



# Variant: CNN\_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

|                  |                 |
|------------------|-----------------|
| Trainable params | 68,455          |
| LOUO mean ± std  | 0.9248 ± 0.0633 |
| Folds run        | 13              |
| Inner-val mode   | user            |
| Per-trial norm   | zscore          |

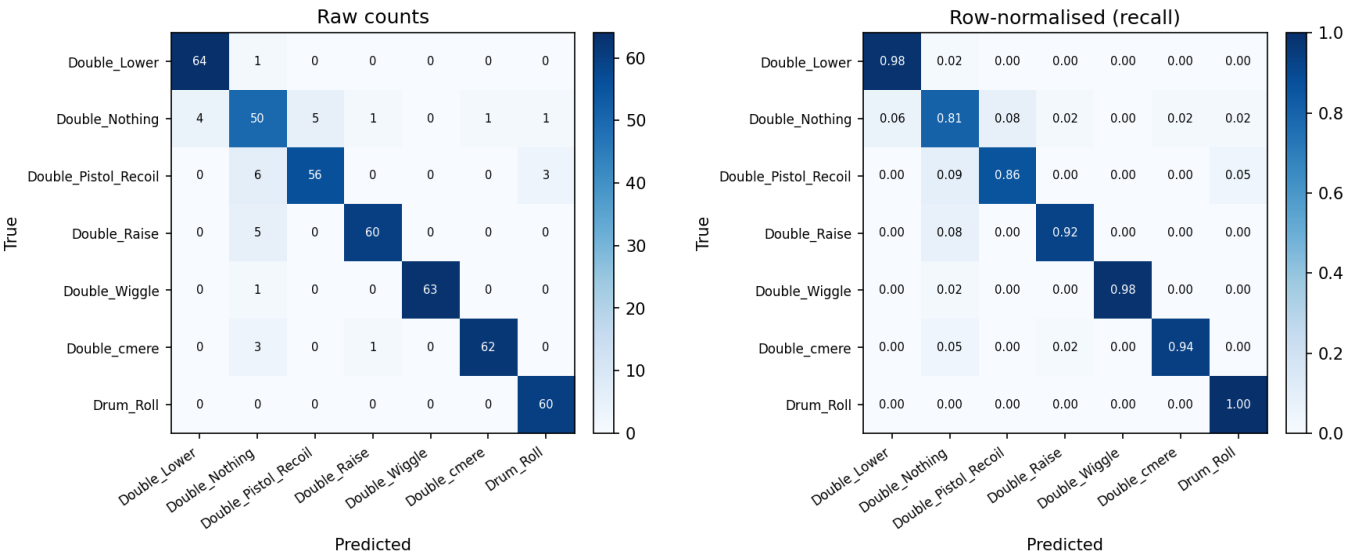
## LOUO per-fold results

| Fold | Held-out user | Inner-val user | n_train | n_test | Accuracy |
|------|---------------|----------------|---------|--------|----------|
| 1    | 11_Mansh      | 12_Marcus      | 377     | 35     | 0.8857   |
| 2    | 12_Marcus     | 14_StephenV2   | 376     | 35     | 0.9714   |
| 3    | 14_StephenV2  | 15_Tash        | 376     | 36     | 0.9444   |
| 4    | 15_Tash       | 16_Bella       | 376     | 35     | 0.9143   |
| 5    | 16_Bella      | 17_Raquel      | 376     | 36     | 0.9722   |
| 6    | 17_Raquel     | 18_Bridgette   | 377     | 35     | 1.0000   |
| 7    | 18_Bridgette  | 1_Alan         | 377     | 35     | 0.9429   |
| 8    | 1_Alan        | 2_Alex         | 377     | 35     | 0.8286   |
| 9    | 2_Alex        | 3_Anghad       | 376     | 35     | 0.9429   |
| 10   | 3_Anghad      | 4_Daniel       | 376     | 36     | 1.0000   |
| 11   | 4_Daniel      | 6_Harry        | 377     | 35     | 0.9714   |
| 12   | 6_Harry       | 8_Jestin       | 388     | 35     | 0.8571   |
| 13   | 8_Jestin      | 11_Mansh       | 388     | 24     | 0.7917   |

## Per-class metrics (aggregated across folds)

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 0.941     | 0.985  | 0.962 | 65      |
| Double_Nothing       | 0.758     | 0.806  | 0.781 | 62      |
| Double_Pistol_Recoil | 0.918     | 0.862  | 0.889 | 65      |
| Double_Raise         | 0.968     | 0.923  | 0.945 | 65      |
| Double_Wiggle        | 1.000     | 0.984  | 0.992 | 64      |
| Double_cmere         | 0.984     | 0.939  | 0.961 | 66      |
| Drum_Roll            | 0.938     | 1.000  | 0.968 | 60      |
| macro avg            | 0.929     | 0.928  | 0.928 | 447     |
| weighted avg         | 0.930     | 0.928  | 0.929 | 447     |

## Confusion matrix (aggregated across folds)





## Variant: CNN\_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

|                     |                     |
|---------------------|---------------------|
| Trainable params    | 105,575             |
| LOUO mean $\pm$ std | 0.9207 $\pm$ 0.0657 |
| Folds run           | 13                  |
| Inner-val mode      | user                |
| Per-trial norm      | zscore              |

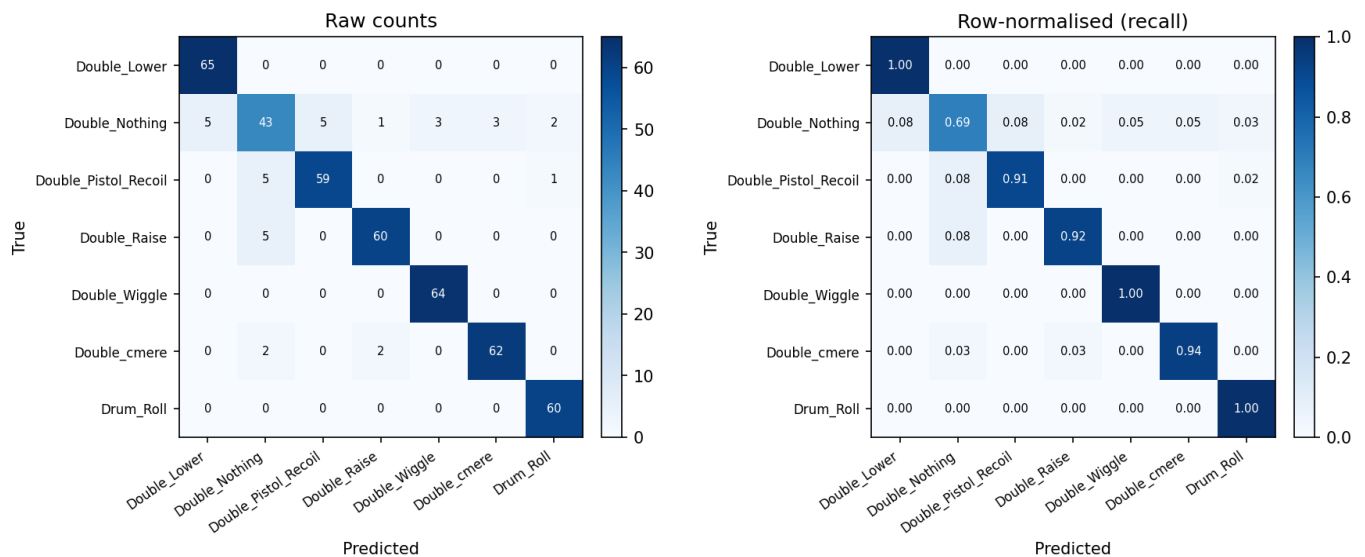
### LOUO per-fold results

| Fold | Held-out user | Inner-val user | n_train | n_test | Accuracy |
|------|---------------|----------------|---------|--------|----------|
| 1    | 11_Mansh      | 12_Marcus      | 377     | 35     | 0.8571   |
| 2    | 12_Marcus     | 14_StephenV2   | 376     | 35     | 1.0000   |
| 3    | 14_StephenV2  | 15_Tash        | 376     | 36     | 0.9444   |
| 4    | 15_Tash       | 16_Bella       | 376     | 35     | 1.0000   |
| 5    | 16_Bella      | 17_Raquel      | 376     | 36     | 0.8889   |
| 6    | 17_Raquel     | 18_Bridgette   | 377     | 35     | 0.9714   |
| 7    | 18_Bridgette  | 1_Alan         | 377     | 35     | 0.9429   |
| 8    | 1_Alan        | 2_Alex         | 377     | 35     | 0.8286   |
| 9    | 2_Alex        | 3_Anghad       | 376     | 35     | 0.9429   |
| 10   | 3_Anghad      | 4_Daniel       | 376     | 36     | 0.9722   |
| 11   | 4_Daniel      | 6_Harry        | 377     | 35     | 0.9714   |
| 12   | 6_Harry       | 8_Jestin       | 388     | 35     | 0.8571   |
| 13   | 8_Jestin      | 11_Mansh       | 388     | 24     | 0.7917   |

### Per-class metrics (aggregated across folds)

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 0.929     | 1.000  | 0.963 | 65      |
| Double_Nothing       | 0.782     | 0.694  | 0.735 | 62      |
| Double_Pistol_Recoil | 0.922     | 0.908  | 0.915 | 65      |
| Double_Raise         | 0.952     | 0.923  | 0.938 | 65      |
| Double_Wiggle        | 0.955     | 1.000  | 0.977 | 64      |
| Double_cmere         | 0.954     | 0.939  | 0.947 | 66      |
| Drum_Roll            | 0.952     | 1.000  | 0.976 | 60      |
| macro avg            | 0.921     | 0.923  | 0.921 | 447     |
| weighted avg         | 0.921     | 0.924  | 0.922 | 447     |

### Confusion matrix (aggregated across folds)



# Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

|                  |                 |
|------------------|-----------------|
| Trainable params | 72,935          |
| LOUO mean ± std  | 0.8802 ± 0.0902 |
| Folds run        | 13              |
| Inner-val mode   | user            |
| Per-trial norm   | zscore          |

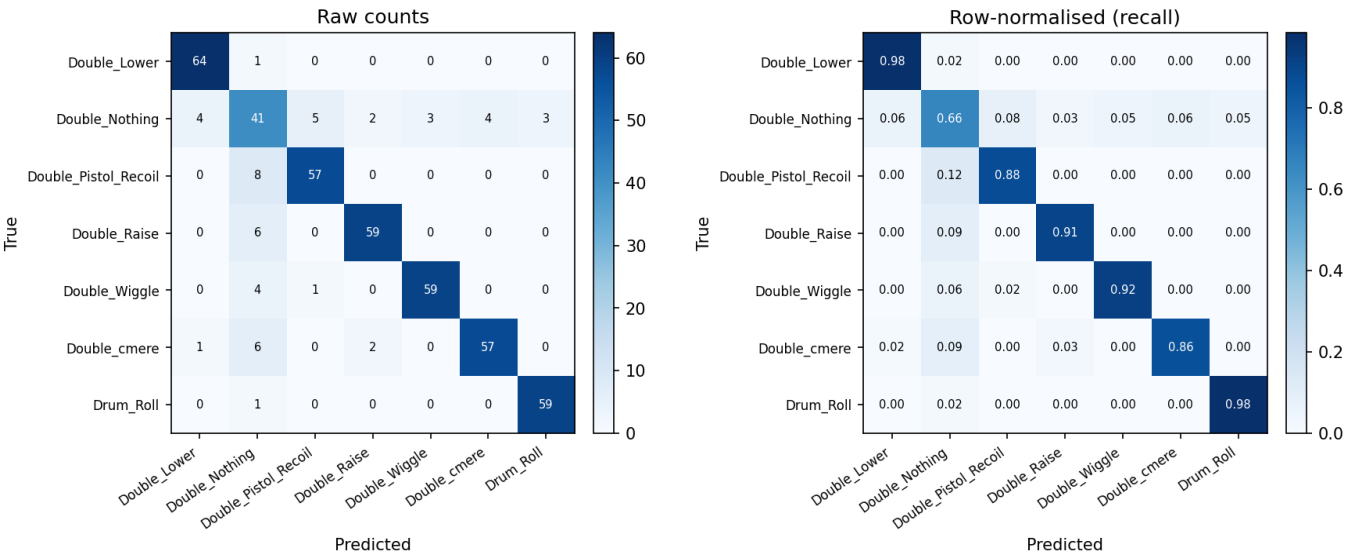
## LOUO per-fold results

| Fold | Held-out user | Inner-val user | n_train | n_test | Accuracy |
|------|---------------|----------------|---------|--------|----------|
| 1    | 11_Mansh      | 12_Marcus      | 377     | 35     | 0.8857   |
| 2    | 12_Marcus     | 14_StephenV2   | 376     | 35     | 0.8571   |
| 3    | 14_StephenV2  | 15_Tash        | 376     | 36     | 0.9444   |
| 4    | 15_Tash       | 16_Bella       | 376     | 35     | 0.9429   |
| 5    | 16_Bella      | 17_Raquel      | 376     | 36     | 0.9167   |
| 6    | 17_Raquel     | 18_Bridgette   | 377     | 35     | 0.9429   |
| 7    | 18_Bridgette  | 1_Alan         | 377     | 35     | 0.8857   |
| 8    | 1_Alan        | 2_Alex         | 377     | 35     | 0.7143   |
| 9    | 2_Alex        | 3_Anghad       | 376     | 35     | 0.8571   |
| 10   | 3_Anghad      | 4_Daniel       | 376     | 36     | 1.0000   |
| 11   | 4_Daniel      | 6_Harry        | 377     | 35     | 0.9429   |
| 12   | 6_Harry       | 8_Jestin       | 388     | 35     | 0.8857   |
| 13   | 8_Jestin      | 11_Mansh       | 388     | 24     | 0.6667   |

## Per-class metrics (aggregated across folds)

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 0.928     | 0.985  | 0.955 | 65      |
| Double_Nothing       | 0.612     | 0.661  | 0.636 | 62      |
| Double_Pistol_Recoil | 0.905     | 0.877  | 0.891 | 65      |
| Double_Raise         | 0.937     | 0.908  | 0.922 | 65      |
| Double_Wiggle        | 0.952     | 0.922  | 0.937 | 64      |
| Double_cmere         | 0.934     | 0.864  | 0.898 | 66      |
| Drum_Roll            | 0.952     | 0.983  | 0.967 | 60      |
| macro avg            | 0.888     | 0.886  | 0.886 | 447     |
| weighted avg         | 0.889     | 0.886  | 0.887 | 447     |

## Confusion matrix (aggregated across folds)



# Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

|                  |                 |
|------------------|-----------------|
| Trainable params | 51,815          |
| LOUO mean ± std  | 0.8771 ± 0.0968 |
| Folds run        | 13              |
| Inner-val mode   | user            |
| Per-trial norm   | zscore          |

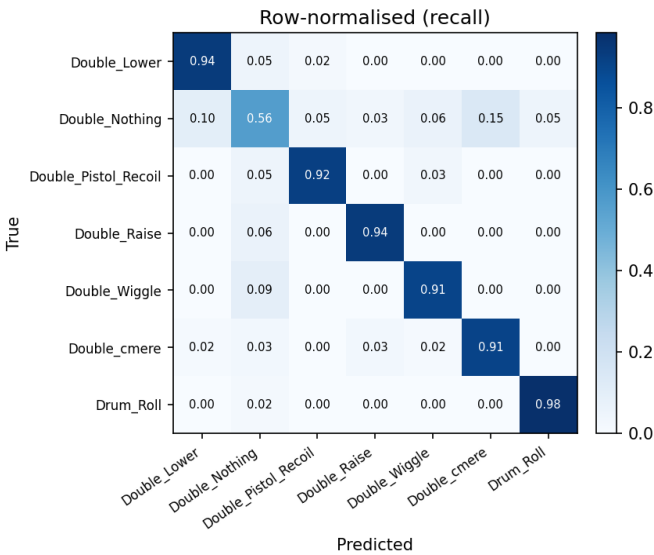
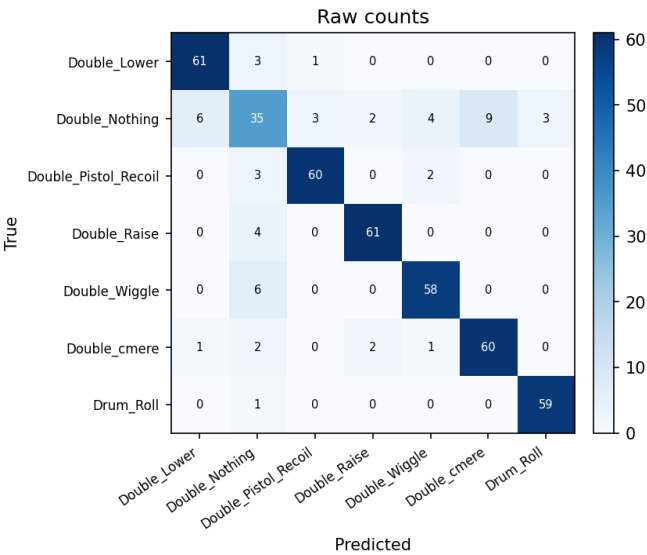
## LOUO per-fold results

| Fold | Held-out user | Inner-val user | n_train | n_test | Accuracy |
|------|---------------|----------------|---------|--------|----------|
| 1    | 11_Mansh      | 12_Marcus      | 377     | 35     | 0.8286   |
| 2    | 12_Marcus     | 14_StephenV2   | 376     | 35     | 0.8857   |
| 3    | 14_StephenV2  | 15_Tash        | 376     | 36     | 0.8889   |
| 4    | 15_Tash       | 16_Bella       | 376     | 35     | 1.0000   |
| 5    | 16_Bella      | 17_Raquel      | 376     | 36     | 0.8611   |
| 6    | 17_Raquel     | 18_Bridgette   | 377     | 35     | 1.0000   |
| 7    | 18_Bridgette  | 1_Alan         | 377     | 35     | 0.9429   |
| 8    | 1_Alan        | 2_Alex         | 377     | 35     | 0.6571   |
| 9    | 2_Alex        | 3_Anghad       | 376     | 35     | 0.8571   |
| 10   | 3_Anghad      | 4_Daniel       | 376     | 36     | 0.9444   |
| 11   | 4_Daniel      | 6_Harry        | 377     | 35     | 0.9143   |
| 12   | 6_Harry       | 8_Jestin       | 388     | 35     | 0.9143   |
| 13   | 8_Jestin      | 11_Mansh       | 388     | 24     | 0.7083   |

## Per-class metrics (aggregated across folds)

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 0.897     | 0.938  | 0.917 | 65      |
| Double_Nothing       | 0.648     | 0.565  | 0.603 | 62      |
| Double_Pistol_Recoil | 0.938     | 0.923  | 0.930 | 65      |
| Double_Raise         | 0.938     | 0.938  | 0.938 | 65      |
| Double_Wiggle        | 0.892     | 0.906  | 0.899 | 64      |
| Double_cmere         | 0.870     | 0.909  | 0.889 | 66      |
| Drum_Roll            | 0.952     | 0.983  | 0.967 | 60      |
| macro avg            | 0.876     | 0.880  | 0.878 | 447     |
| weighted avg         | 0.877     | 0.881  | 0.879 | 447     |

## Confusion matrix (aggregated across folds)



# Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

|                  |                 |
|------------------|-----------------|
| Trainable params | 70,247          |
| LOUO mean ± std  | 0.8683 ± 0.0690 |
| Folds run        | 13              |
| Inner-val mode   | user            |
| Per-trial norm   | zscore          |

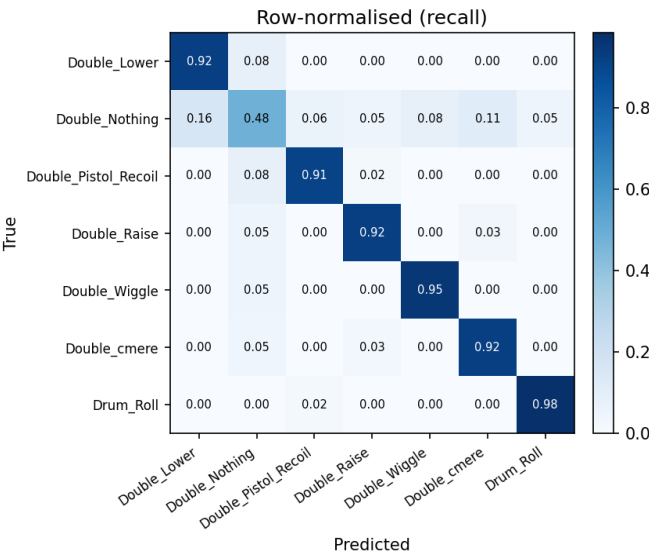
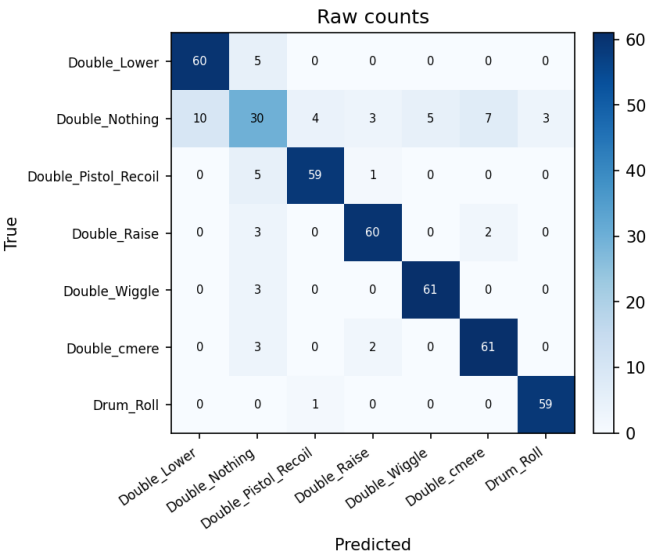
## LOUO per-fold results

| Fold | Held-out user | Inner-val user | n_train | n_test | Accuracy |
|------|---------------|----------------|---------|--------|----------|
| 1    | 11_Mansh      | 12_Marcus      | 377     | 35     | 0.8000   |
| 2    | 12_Marcus     | 14_StephenV2   | 376     | 35     | 0.9143   |
| 3    | 14_StephenV2  | 15_Tash        | 376     | 36     | 0.8611   |
| 4    | 15_Tash       | 16_Bella       | 376     | 35     | 0.9143   |
| 5    | 16_Bella      | 17_Raquel      | 376     | 36     | 0.9167   |
| 6    | 17_Raquel     | 18_Bridgette   | 377     | 35     | 0.9429   |
| 7    | 18_Bridgette  | 1_Alan         | 377     | 35     | 0.9429   |
| 8    | 1_Alan        | 2_Alex         | 377     | 35     | 0.7714   |
| 9    | 2_Alex        | 3_Anghad       | 376     | 35     | 0.8857   |
| 10   | 3_Anghad      | 4_Daniel       | 376     | 36     | 0.9167   |
| 11   | 4_Daniel      | 6_Harry        | 377     | 35     | 0.8857   |
| 12   | 6_Harry       | 8_Jestin       | 388     | 35     | 0.8286   |
| 13   | 8_Jestin      | 11_Mansh       | 388     | 24     | 0.7083   |

## Per-class metrics (aggregated across folds)

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 0.857     | 0.923  | 0.889 | 65      |
| Double_Nothing       | 0.612     | 0.484  | 0.541 | 62      |
| Double_Pistol_Recoil | 0.922     | 0.908  | 0.915 | 65      |
| Double_Raise         | 0.909     | 0.923  | 0.916 | 65      |
| Double_Wiggle        | 0.924     | 0.953  | 0.938 | 64      |
| Double_cmere         | 0.871     | 0.924  | 0.897 | 66      |
| Drum_Roll            | 0.952     | 0.983  | 0.967 | 60      |
| macro avg            | 0.864     | 0.871  | 0.866 | 447     |
| weighted avg         | 0.865     | 0.872  | 0.867 | 447     |

## Confusion matrix (aggregated across folds)



## Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

|                     |                     |
|---------------------|---------------------|
| Trainable params    | 41,767              |
| LOUO mean $\pm$ std | 0.8311 $\pm$ 0.0742 |
| Folds run           | 13                  |
| Inner-val mode      | user                |
| Per-trial norm      | zscore              |

### LOUO per-fold results

| Fold | Held-out user | Inner-val user | n_train | n_test | Accuracy |
|------|---------------|----------------|---------|--------|----------|
| 1    | 11_Mansh      | 12_Marcus      | 377     | 35     | 0.8286   |
| 2    | 12_Marcus     | 14_StephenV2   | 376     | 35     | 0.8286   |
| 3    | 14_StephenV2  | 15_Tash        | 376     | 36     | 0.9167   |
| 4    | 15_Tash       | 16_Bella       | 376     | 35     | 0.8857   |
| 5    | 16_Bella      | 17_Raquel      | 376     | 36     | 0.8611   |
| 6    | 17_Raquel     | 18_Bridgette   | 377     | 35     | 0.9143   |
| 7    | 18_Bridgette  | 1_Alan         | 377     | 35     | 0.8571   |
| 8    | 1_Alan        | 2_Alex         | 377     | 35     | 0.6571   |
| 9    | 2_Alex        | 3_Anghad       | 376     | 35     | 0.7714   |
| 10   | 3_Anghad      | 4_Daniel       | 376     | 36     | 0.8611   |
| 11   | 4_Daniel      | 6_Harry        | 377     | 35     | 0.8857   |
| 12   | 6_Harry       | 8_Jestin       | 388     | 35     | 0.8286   |
| 13   | 8_Jestin      | 11_Mansh       | 388     | 24     | 0.7083   |

### Per-class metrics (aggregated across folds)

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 0.871     | 0.938  | 0.904 | 65      |
| Double_Nothing       | 0.500     | 0.387  | 0.436 | 62      |
| Double_Pistol_Recoil | 0.905     | 0.877  | 0.891 | 65      |
| Double_Raise         | 0.884     | 0.938  | 0.910 | 65      |
| Double_Wiggle        | 0.862     | 0.781  | 0.820 | 64      |
| Double_cmere         | 0.845     | 0.909  | 0.876 | 66      |
| Drum_Roll            | 0.882     | 1.000  | 0.938 | 60      |
| macro avg            | 0.821     | 0.833  | 0.825 | 447     |
| weighted avg         | 0.823     | 0.834  | 0.826 | 447     |

### Confusion matrix (aggregated across folds)

